

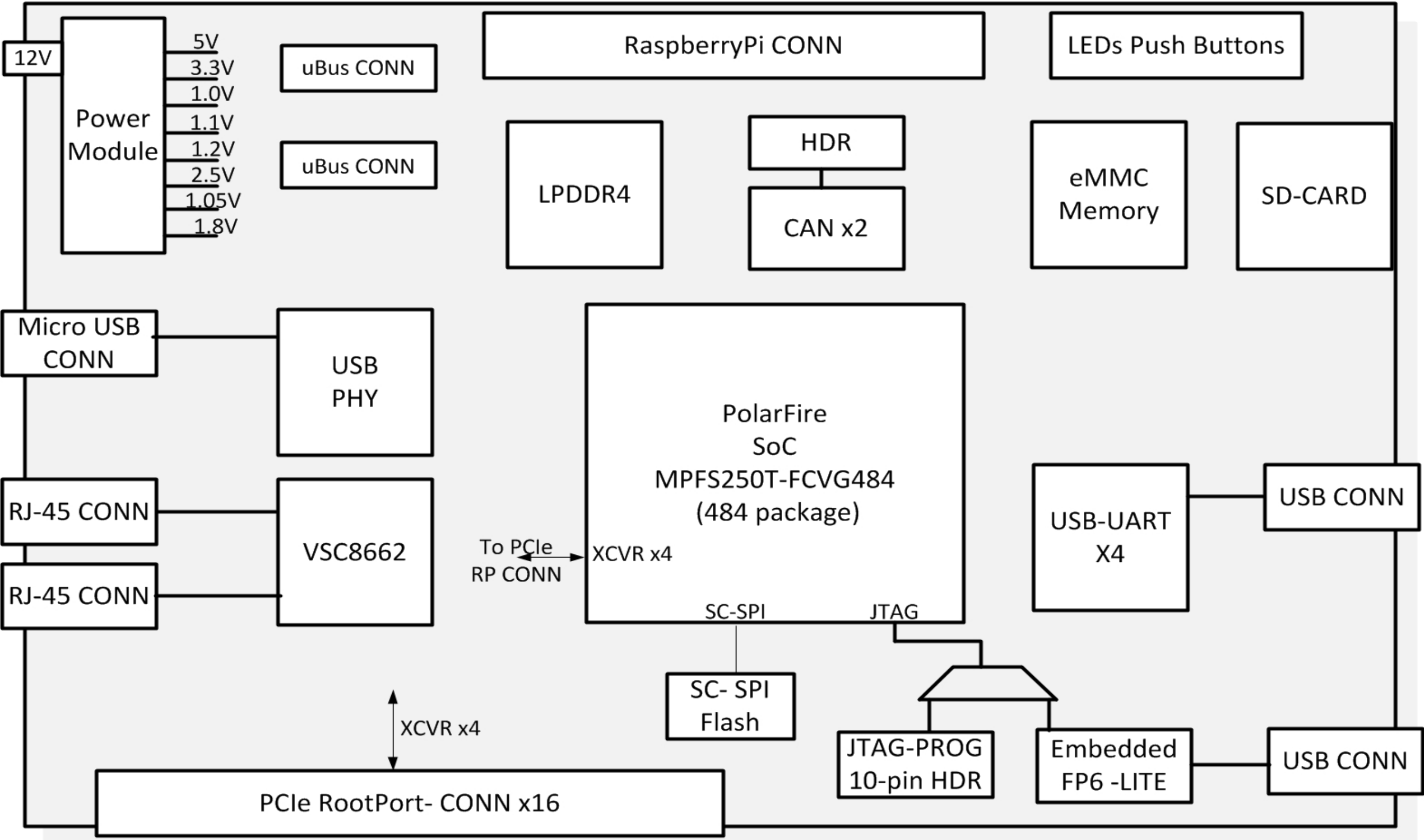
ICICLE Kit Rev1.0

PAGE NO	TITLE	PAGE NO	TITLE
01	TITLE PAGE	18	FP6_DDR3 MEMORY
02	BLOCK DIAGRAM	19	FP6_BANK2,3,4
03	LPDDR4_CONNECTION	20	FP6_BANK-1,5,6,7and MISC
04	VSC8662_MAC	21	FP6_USB3320_IF
05	VSC8662_G5_INTERFACE	22	FP6_POWER &GND
06	VSC8662_POWER & GND	23	POWER BLOCK DIAGRAM
07	PCIE CONNECTORS & MIKROBUS	24	POWER JACK-12V
08	USB-UART INTERFACE	25	POWER-5P0V_1P8V
09	USB INTERFACES-CON1	26	POWER-VDD
10	BANK-3-PROGRAMMING	27	1P1V_LPDDR4 & 1P2V_PHY_VSC
11	eMMC_INTERFACE	28	POWER:3P3V_VDDA
12	CAN & RPI INTERFACE	29	POWER:2P5V, DDR3 & VDDAUX4
13	PolarFire_SoC_Bank_Assign		
14	LEDs & DEBUG CIRCUITRY		
15	G5 SoC POWER		
16	DECOUPLING CAPACITOR		
17	FP6_BANK0-DDR3		



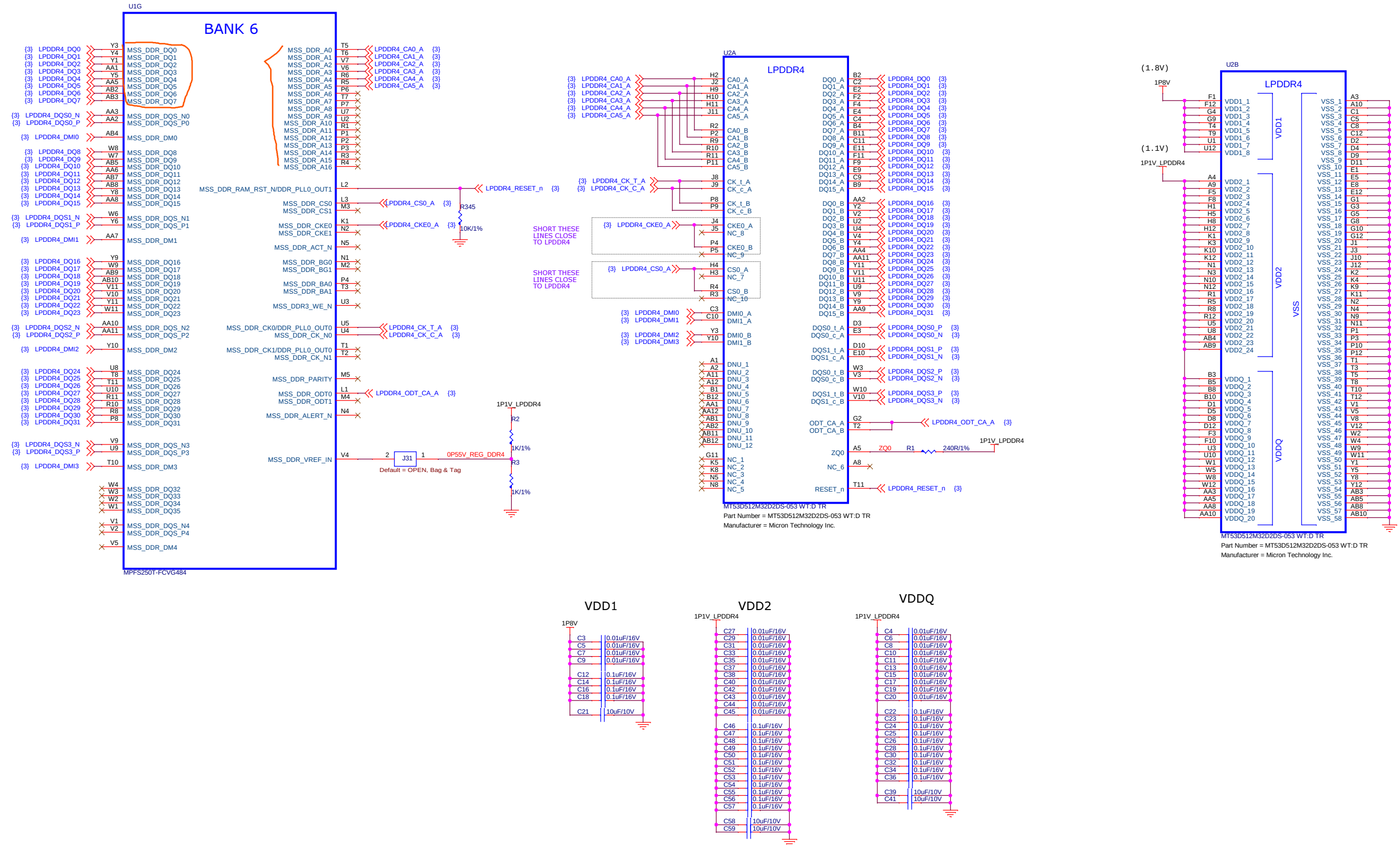
Project Name:			ICICLE Kit		
Rev	Part Number:				PACTRON:
1.0	DVP-102-000536-001				
Originator:					
Size	Date:				Sheet
B	Tuesday, June 23, 2020				1 of 29

ICICLE KIT

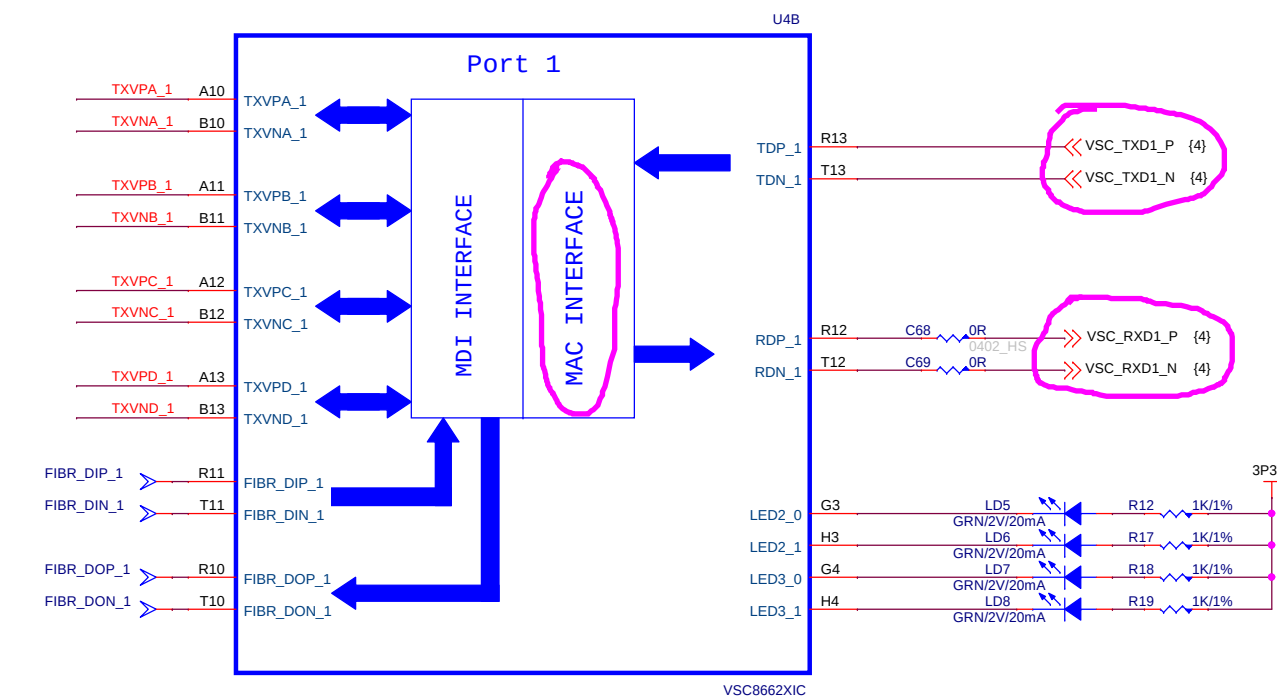
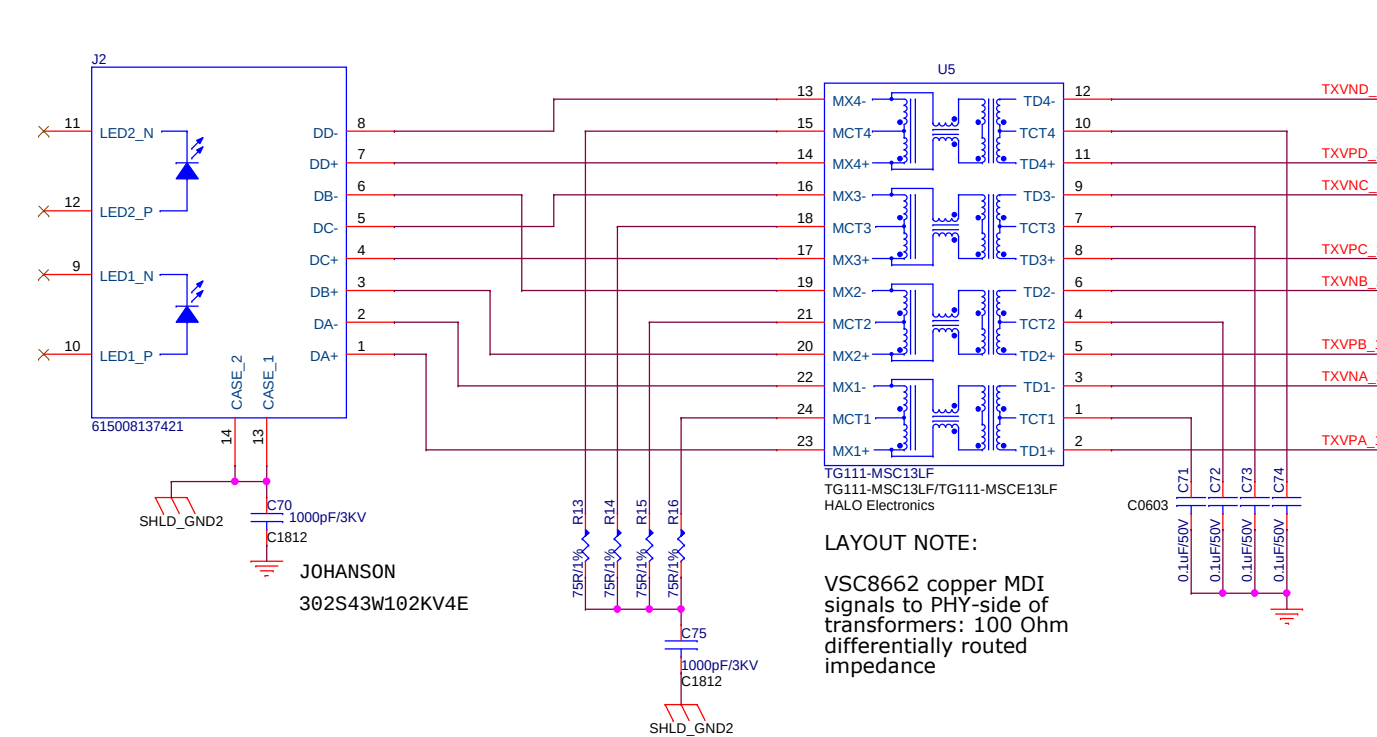
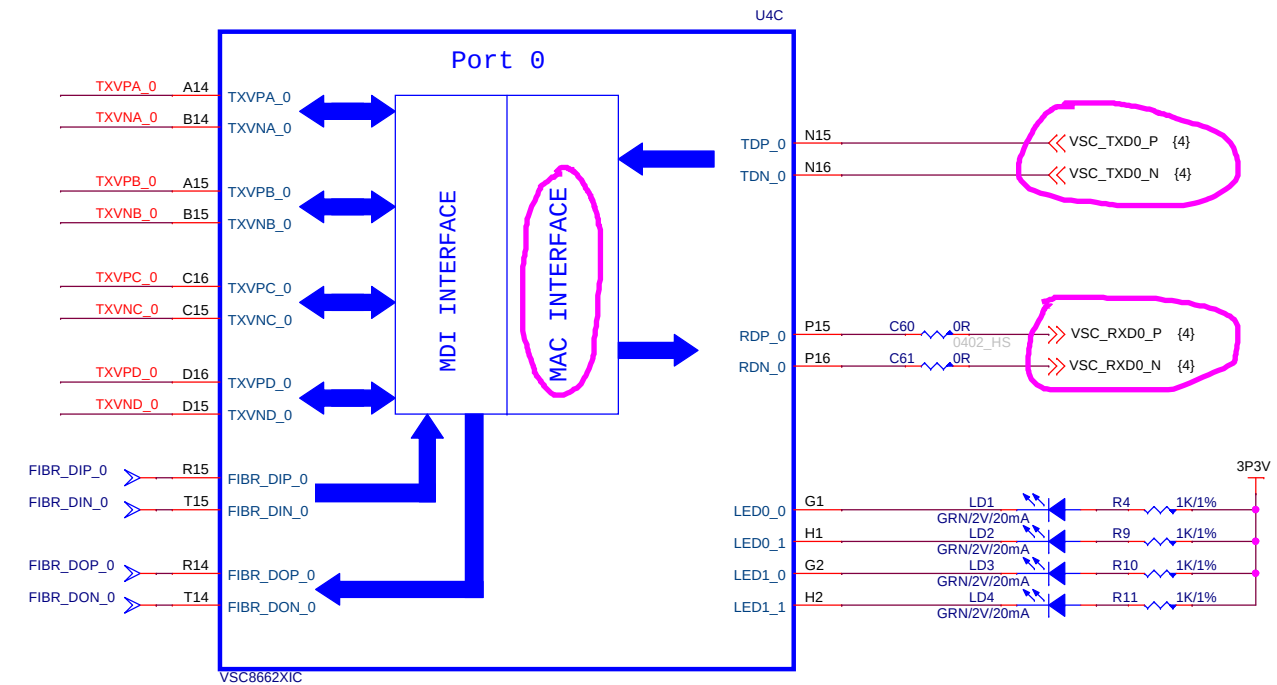
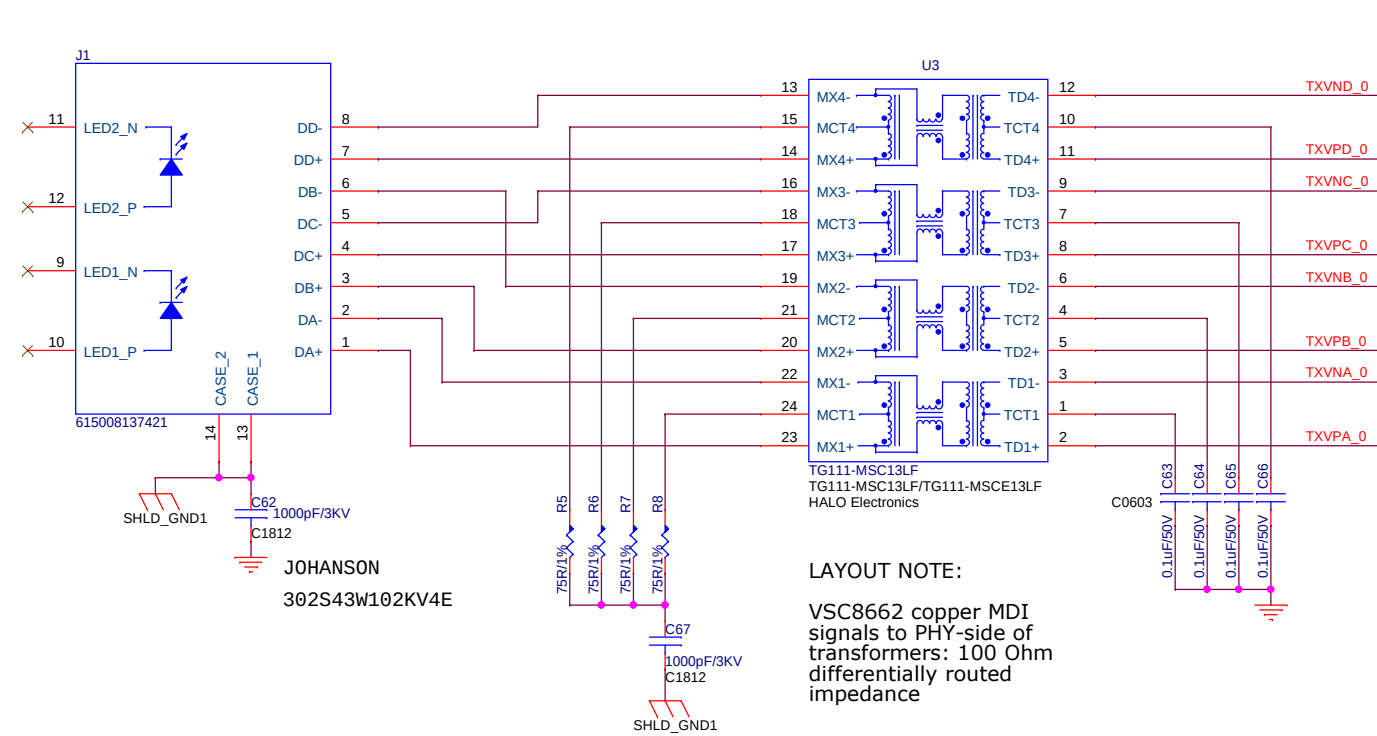


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DATE:	Tuesday, June 23, 2020	SH 2 OF 29

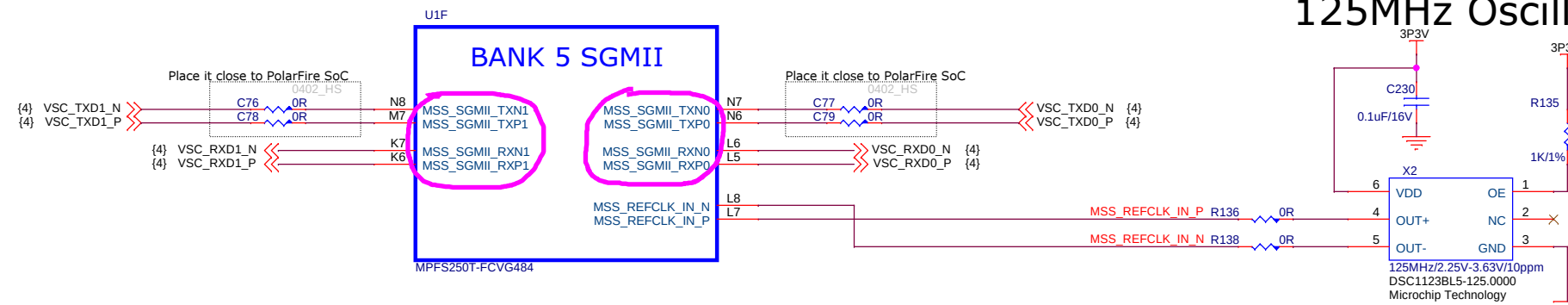
LPDDR4 Connection



VSC8662 - SGMII Interface

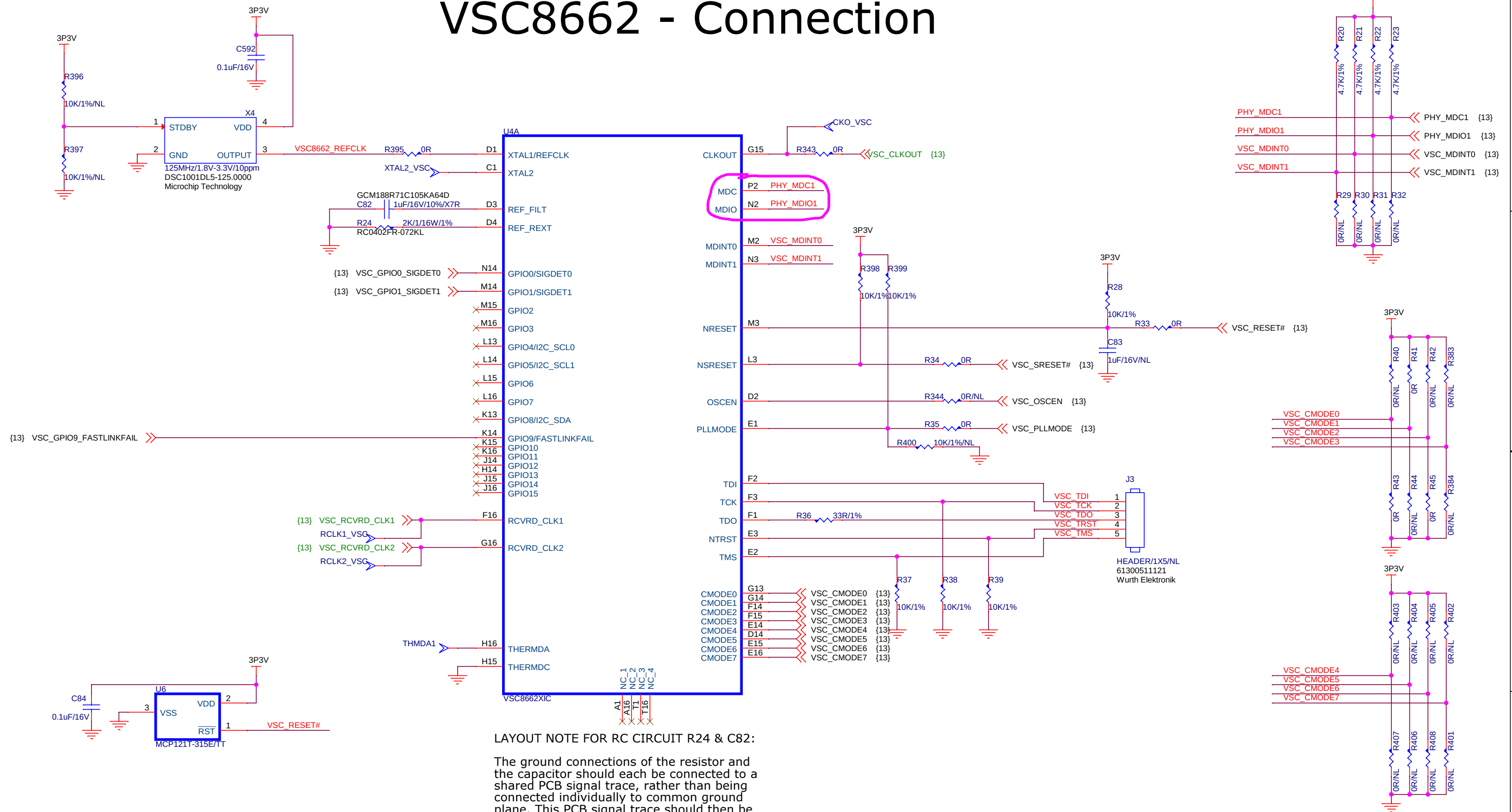


125MHz Oscillator



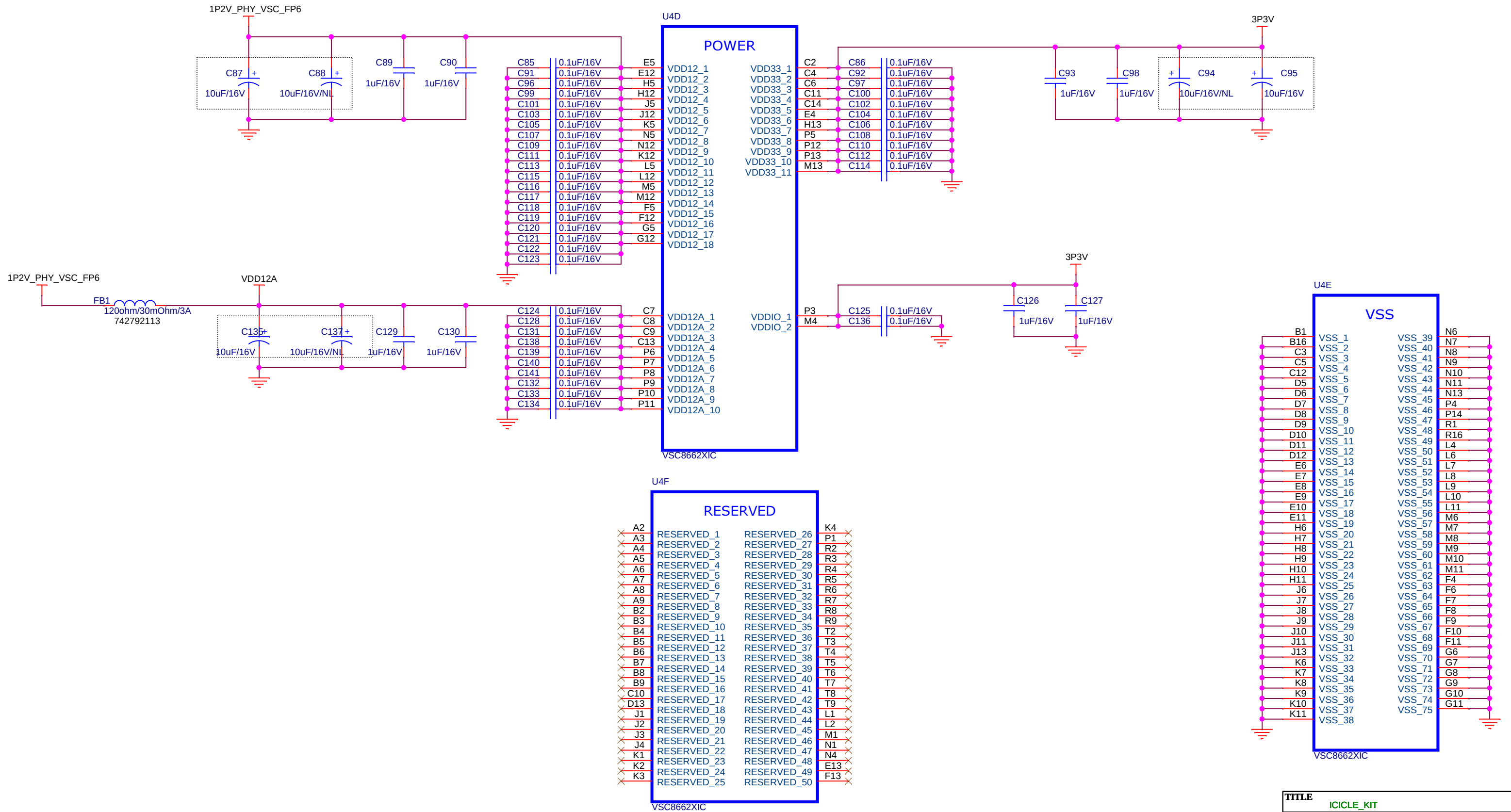
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DATE:	Tuesday, June 23, 2020	SH 4 OF 29

VSC8662 - Connection

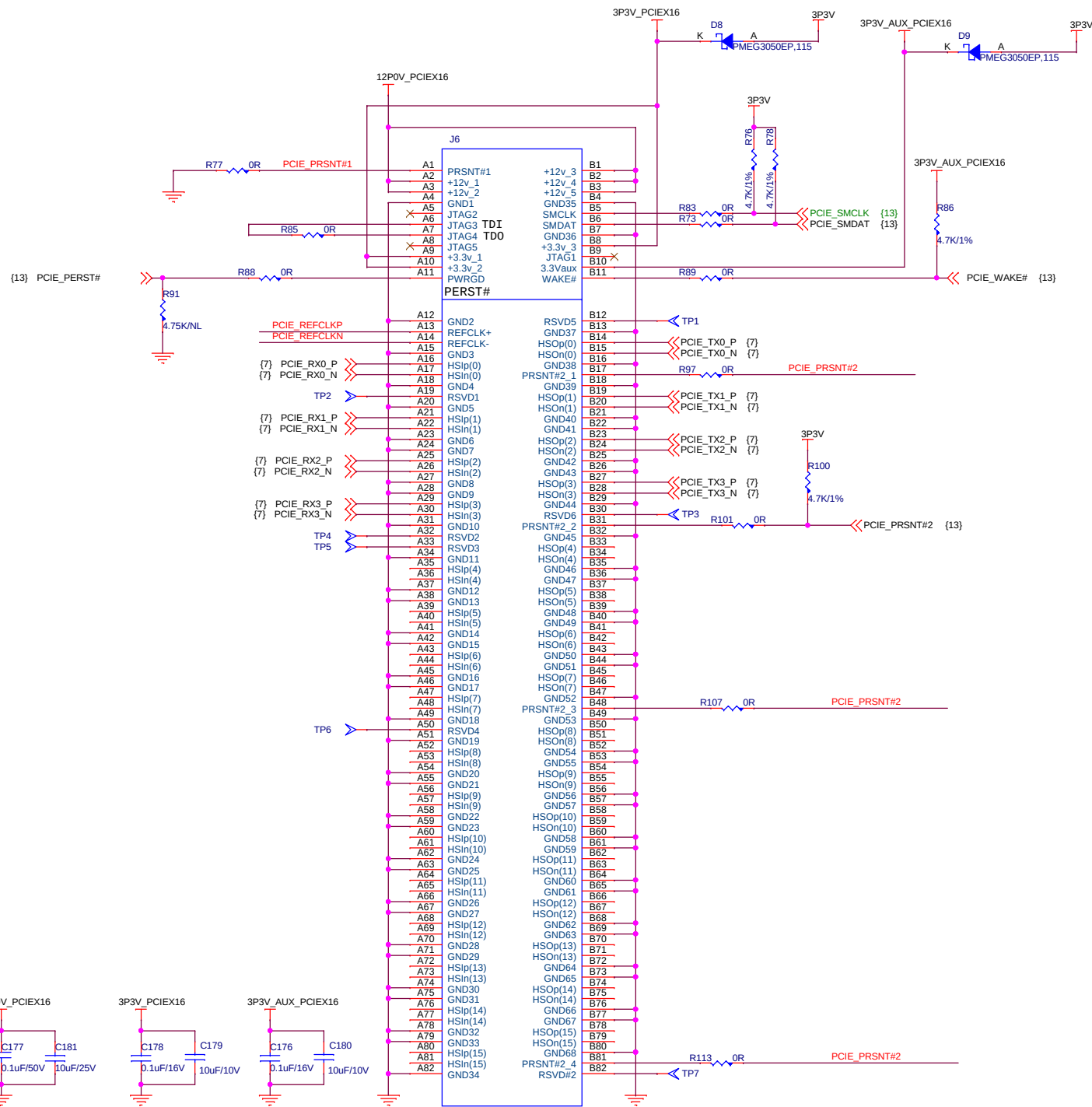


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DATE:	SH 5 OF 29	
Tuesday, June 23, 2020		

VSC8662-Power & Ground

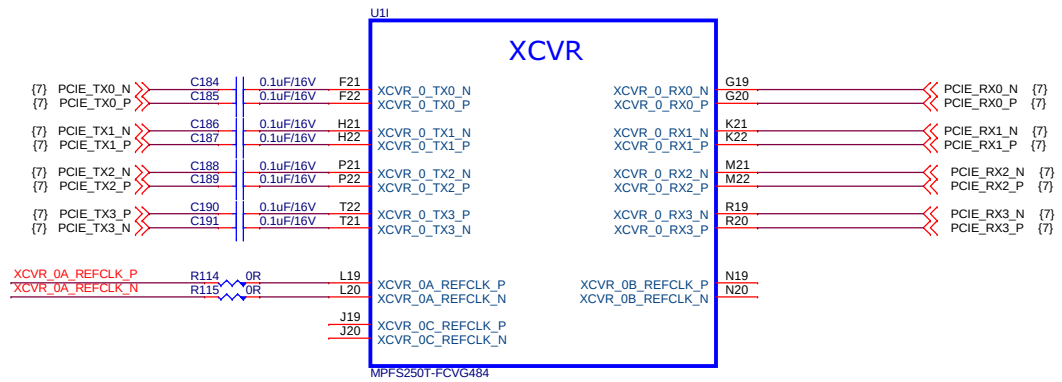
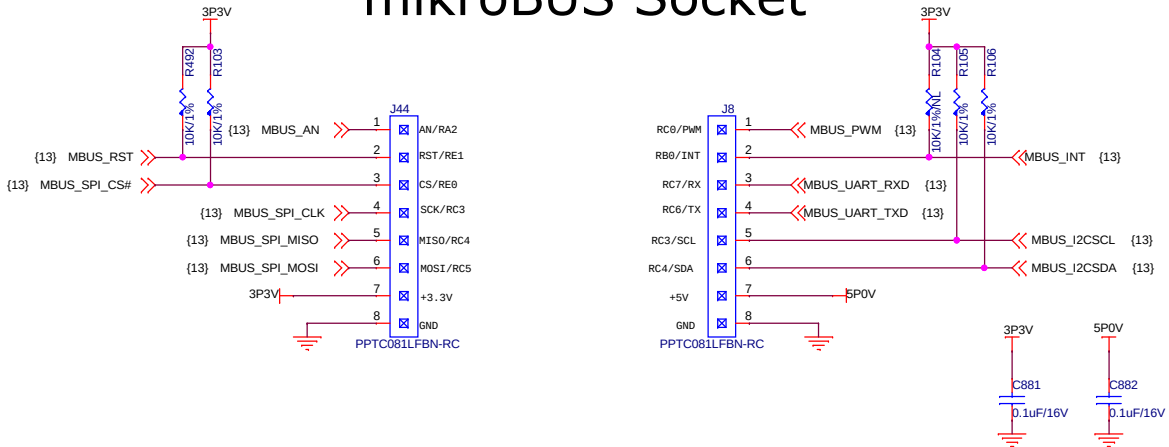


PCIEX16 Connection

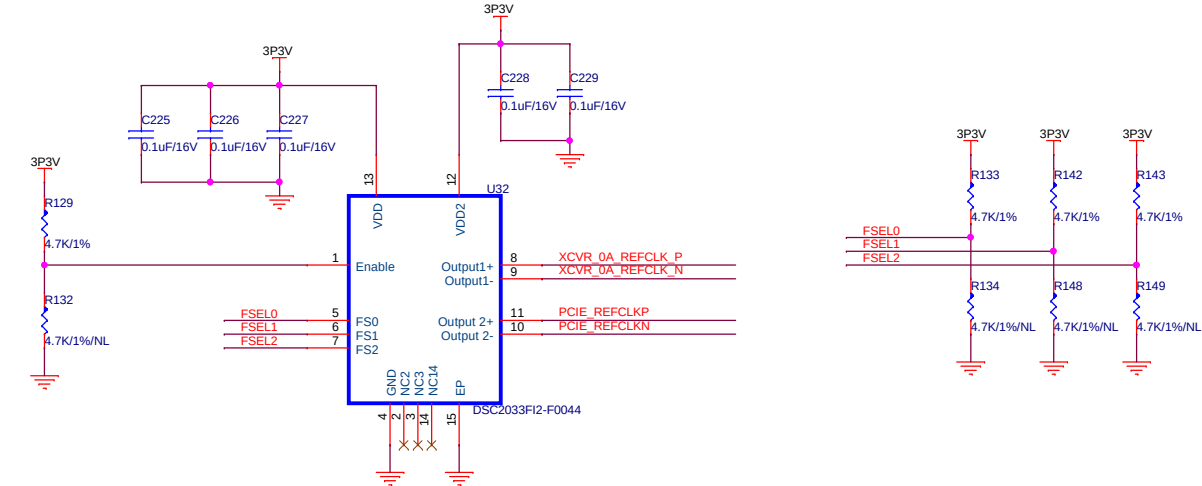


10025026-10103TLF
Part Number = 10025026-10103TLF
Manufacturer = Amphenol ICC (FCI)
Straddle Mount

mikroBUS Socket

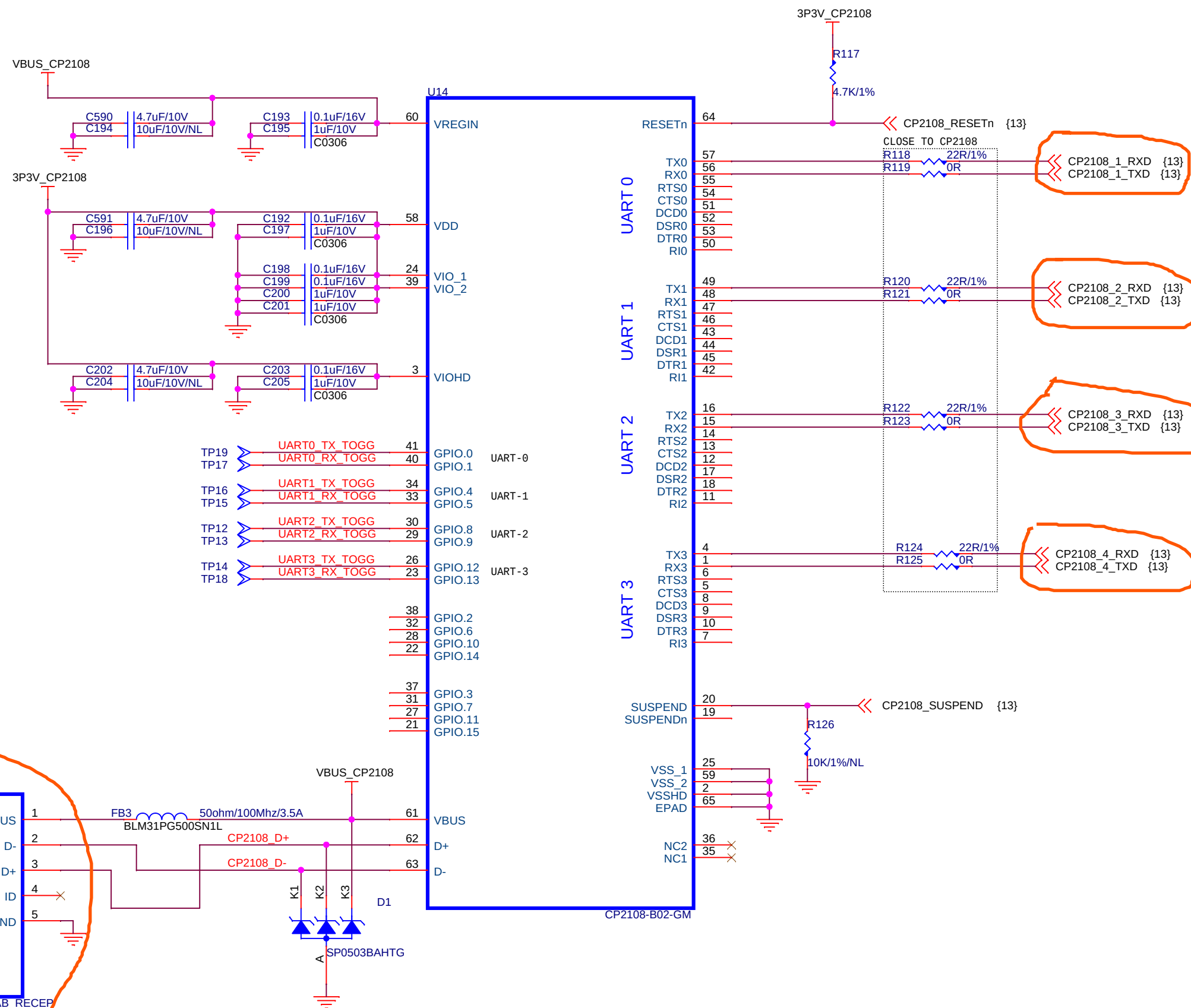


CLOCK BUFFER - 100MHz

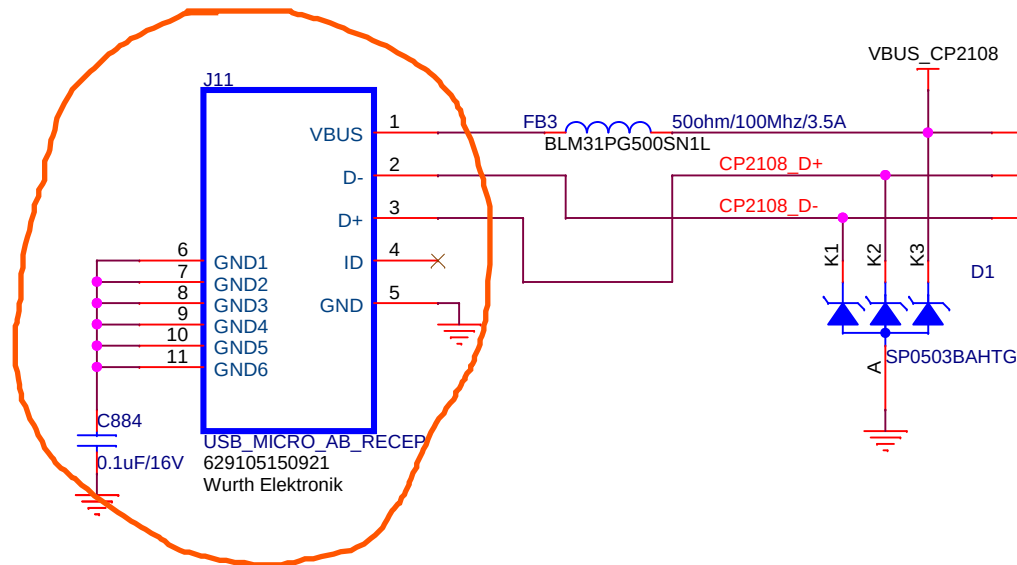


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Custom		1.0
DATE: Tuesday, June 30, 2020		SH 7 OF 29

CP2108 CONNECTION



PolarFire SoC



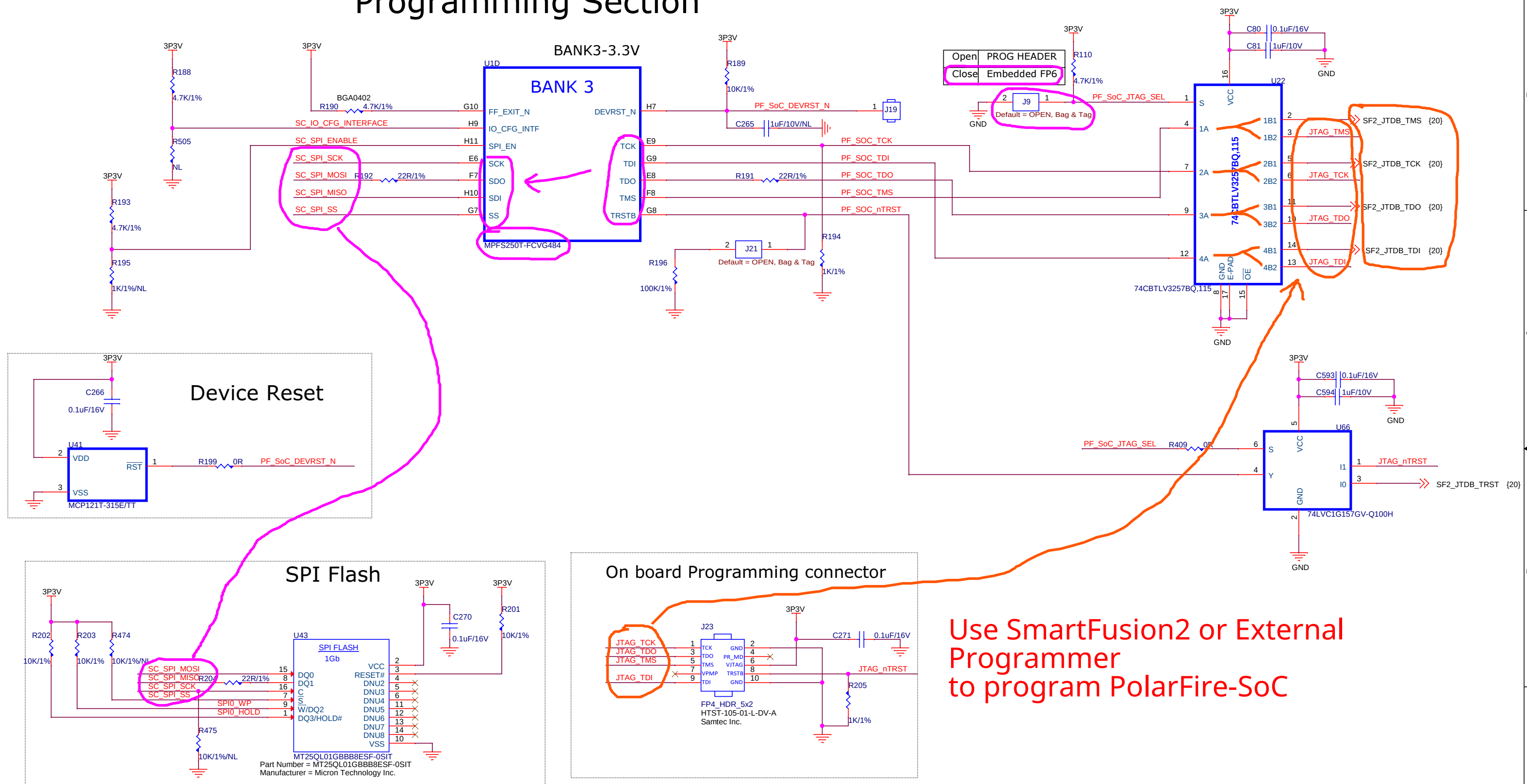
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SIZE B	DOCUMENT NO.		REV 1.0
DATE: Tuesday, June 23, 2020		SH 8 OF 29	

USB3340是Microchip推出的增强型单电源USB 2.0高速ULPI物理层收发器芯片，是USB3320的升级替代型号，主打更低功耗、更完善的工业级防护和充电检测能力。



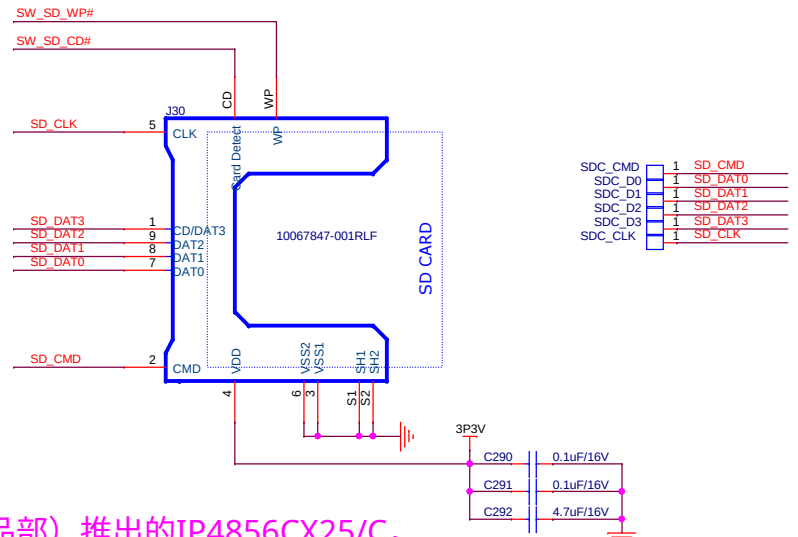
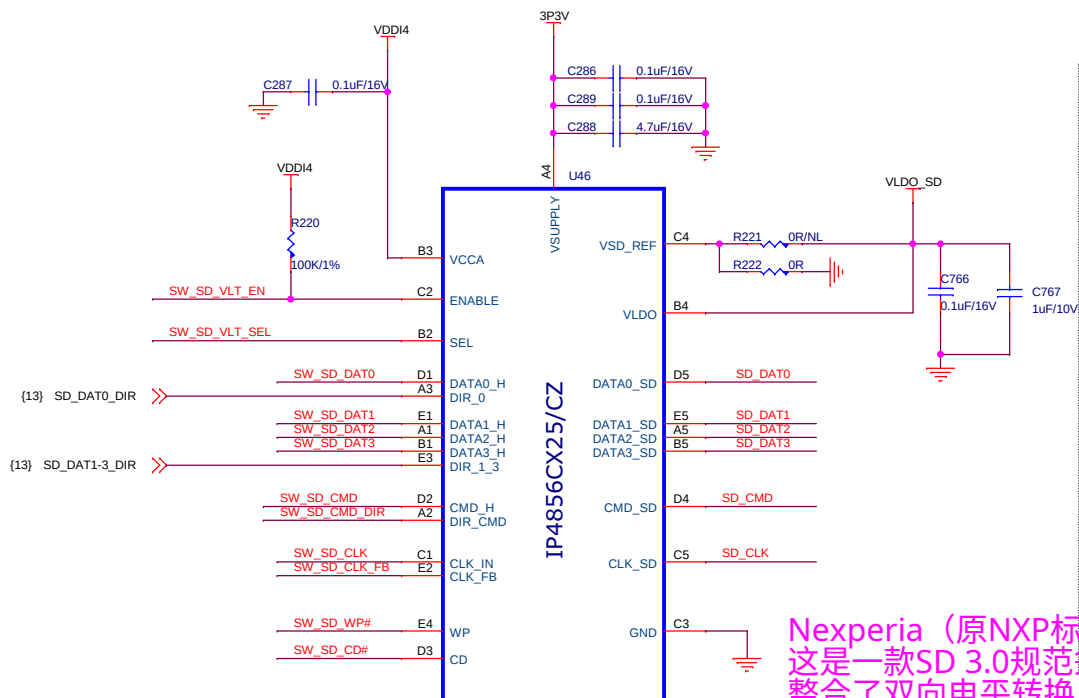
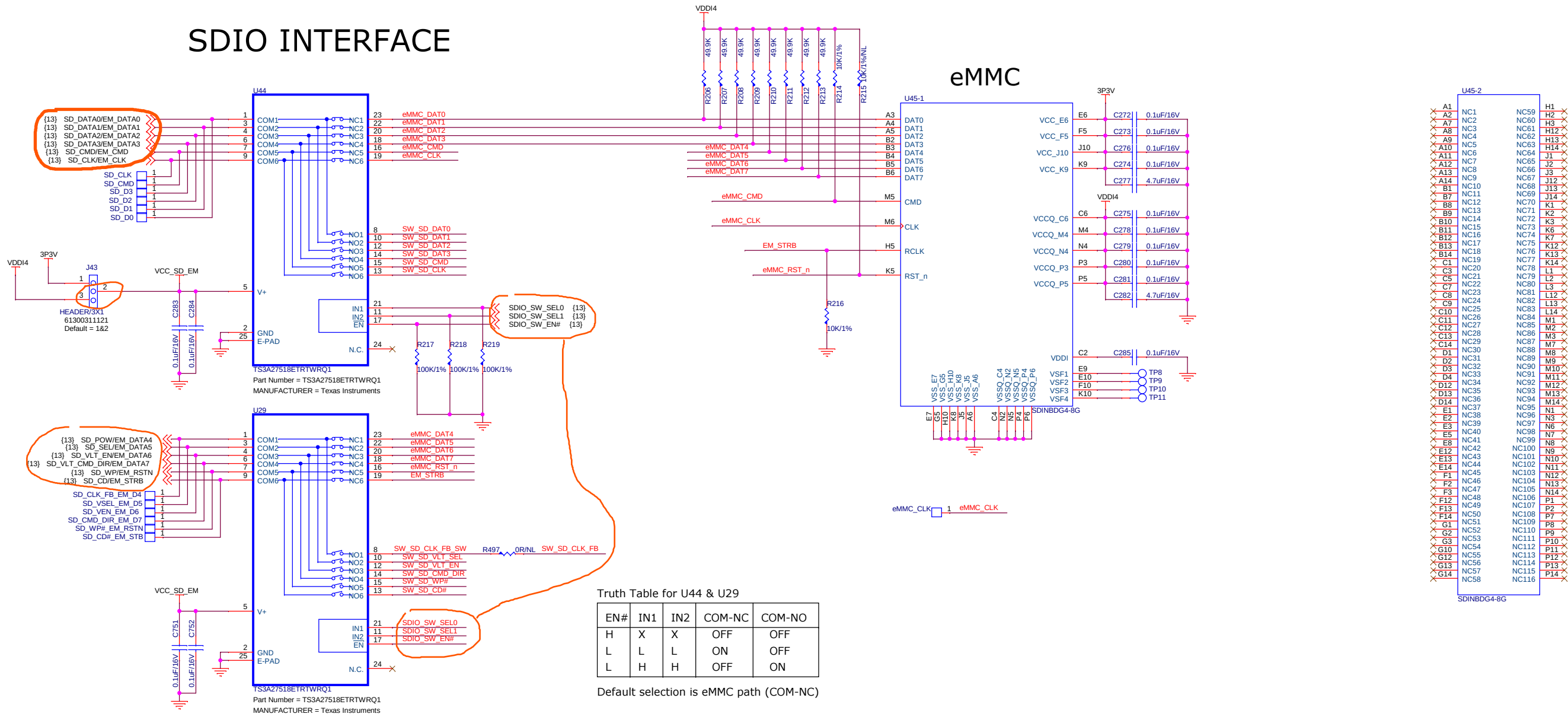
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DATE: Tuesday, June 23, 2020				SH 9	OF 29

Programming Section



Use SmartFusion2 or External Programmer to program PolarFire-SoC

SDIO INTERFACE



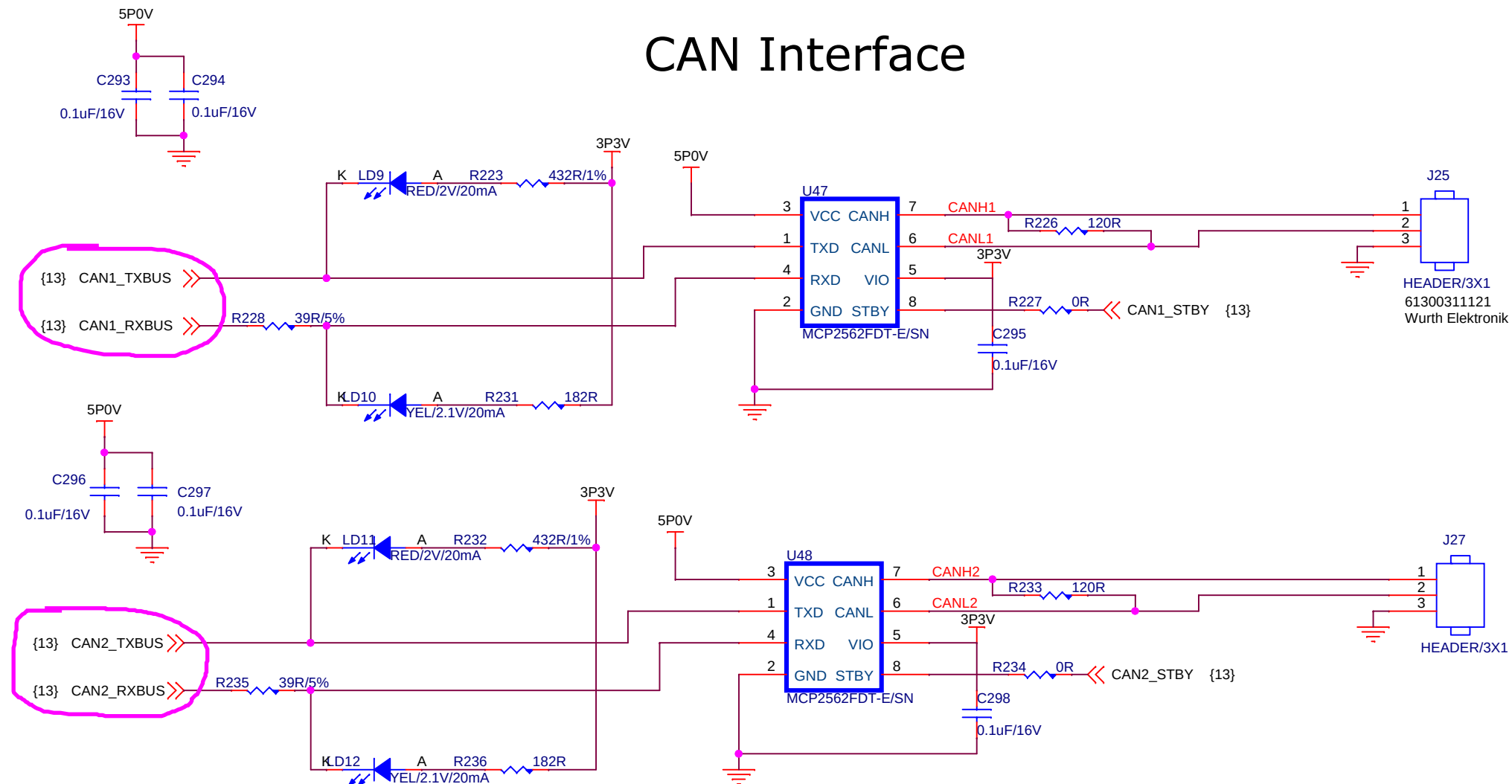
Nexperia (原NXP标准产品部) 推出的IP4856CX25/C, 这是一款SD 3.0规范兼容的集成式SD卡接口芯片, 整合了双向电平转换、EMI滤波和ESD防护三大核心功能

SEL-H- 1.8V Operation
SEL-L- 3V Operation



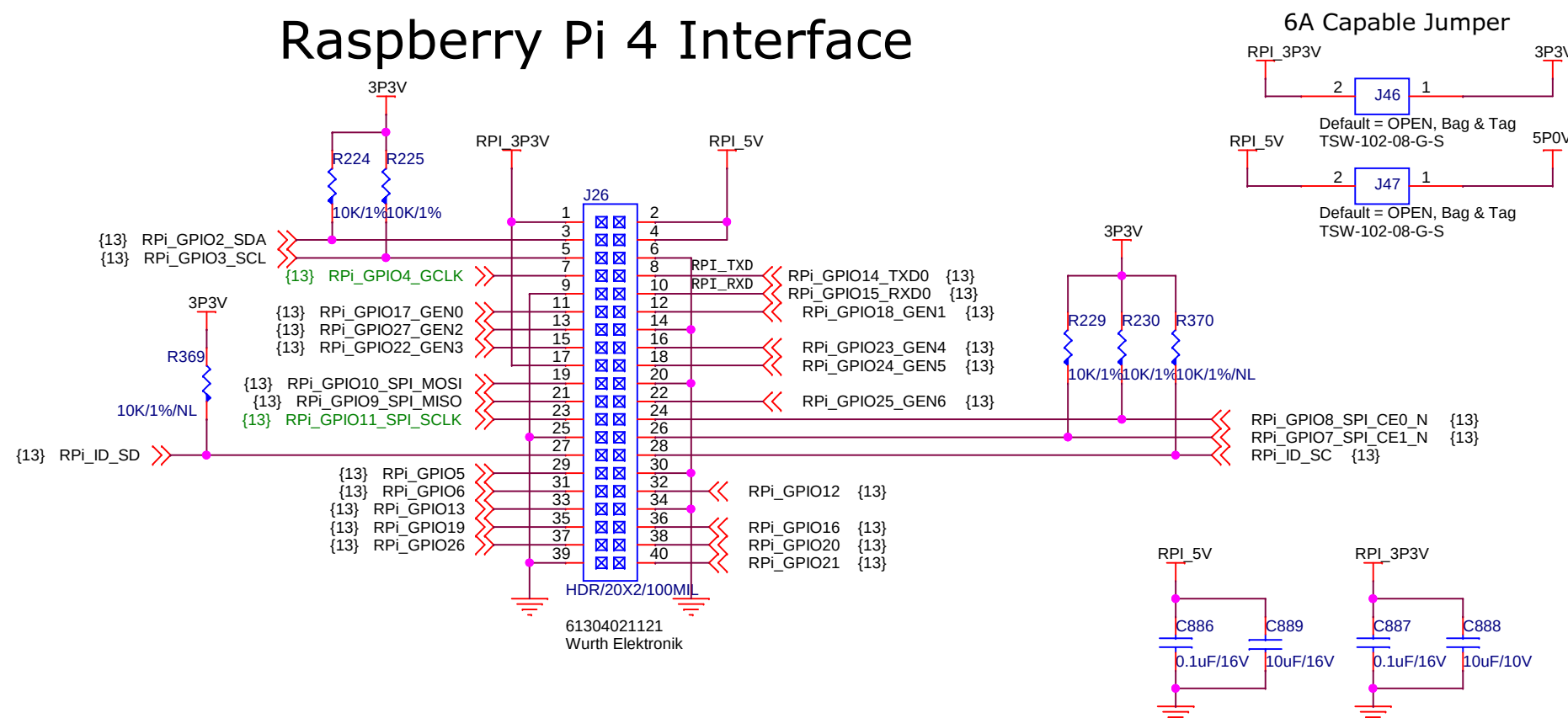
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C		1.0
DATE: Tuesday, June 23, 2020		SH 11 OF 29

CAN Interface



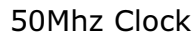
CAN1_TXBUS	R237	68K/1%
CAN1_RXBUS	R238	68K/1%
CAN2_TXBUS	R239	68K/1%
CAN2_RXBUS	R240	68K/1%

Raspberry Pi 4 Interface



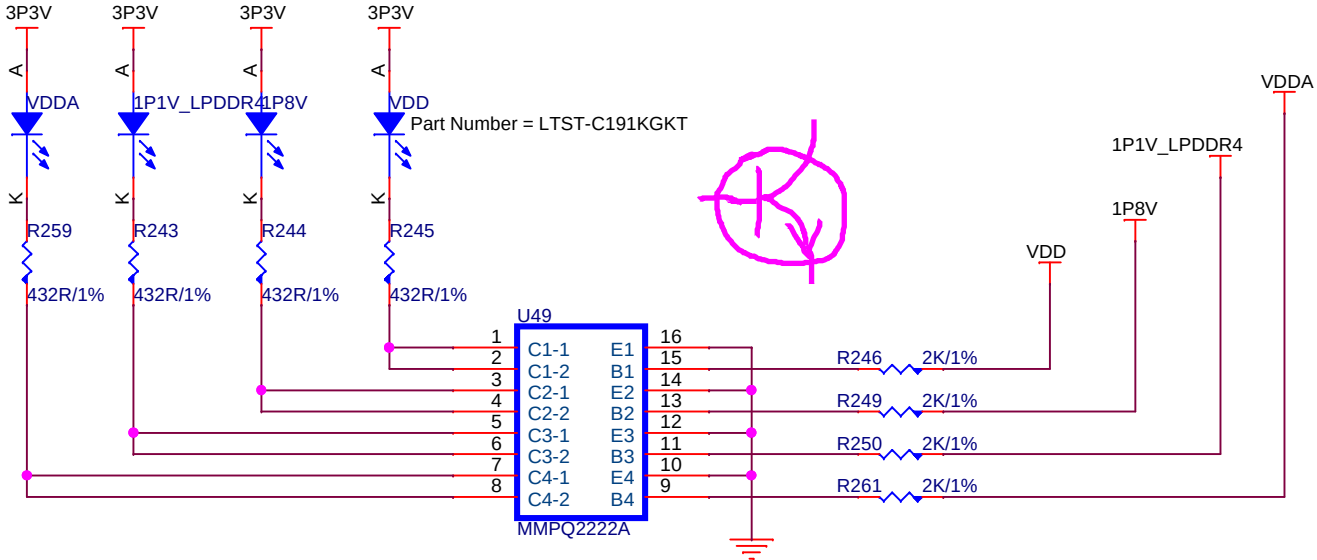
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DATE: Tuesday, June 23, 2020		SH 12	OF 29

PolarFire SoC

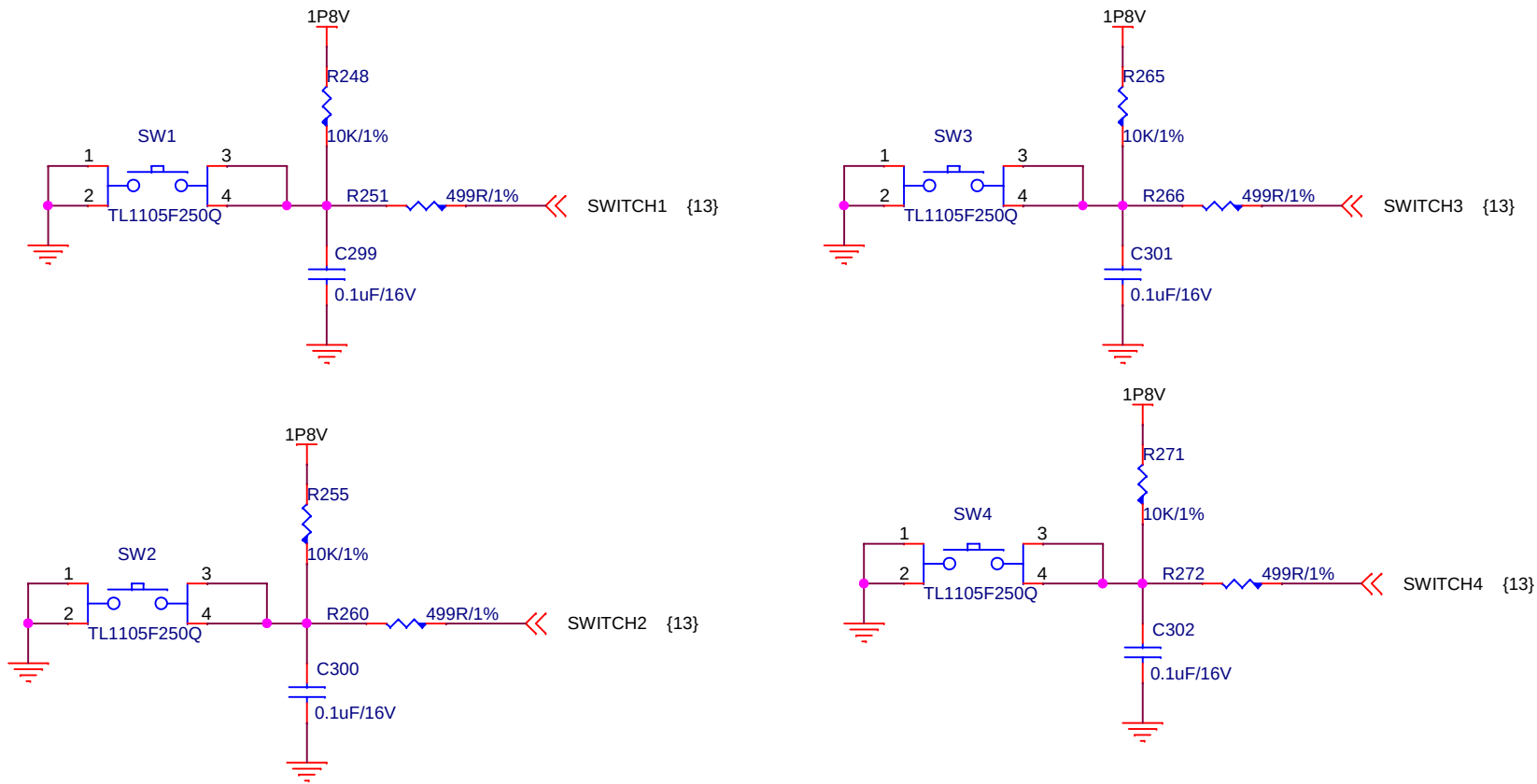


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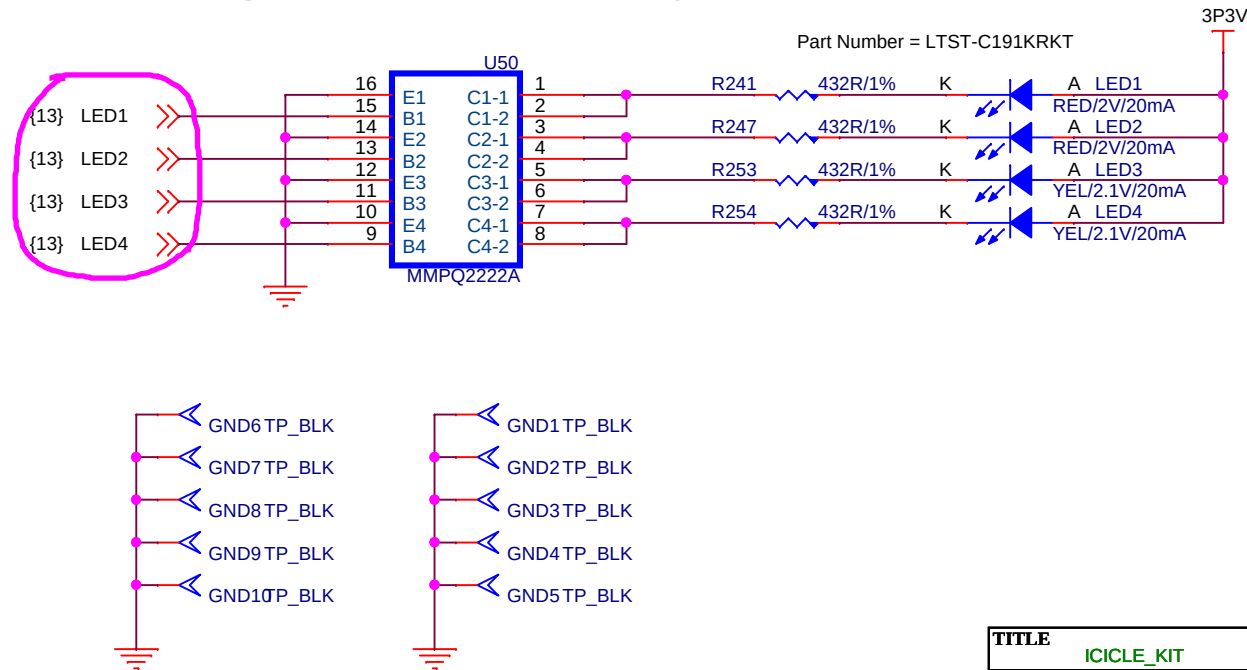
Power LEDs



Debug Circuitry

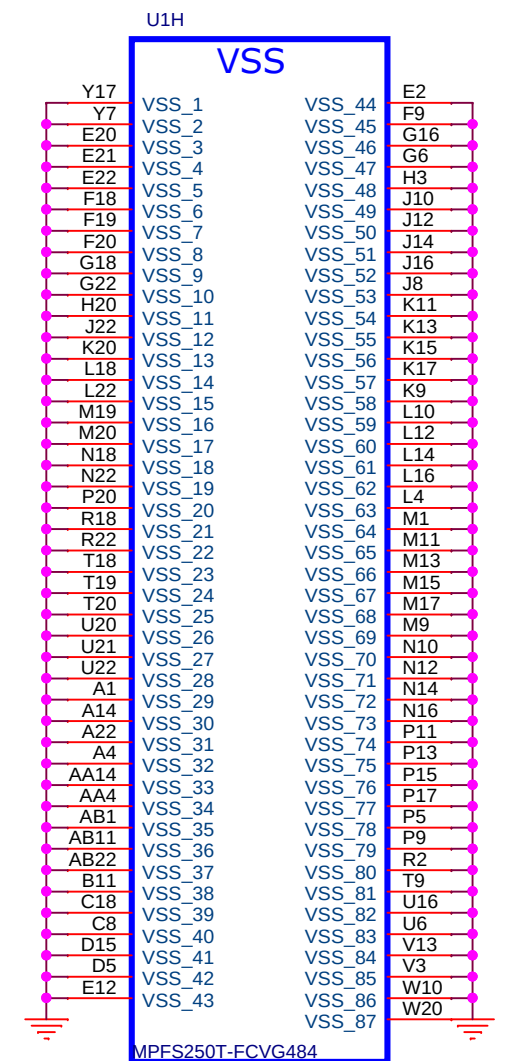
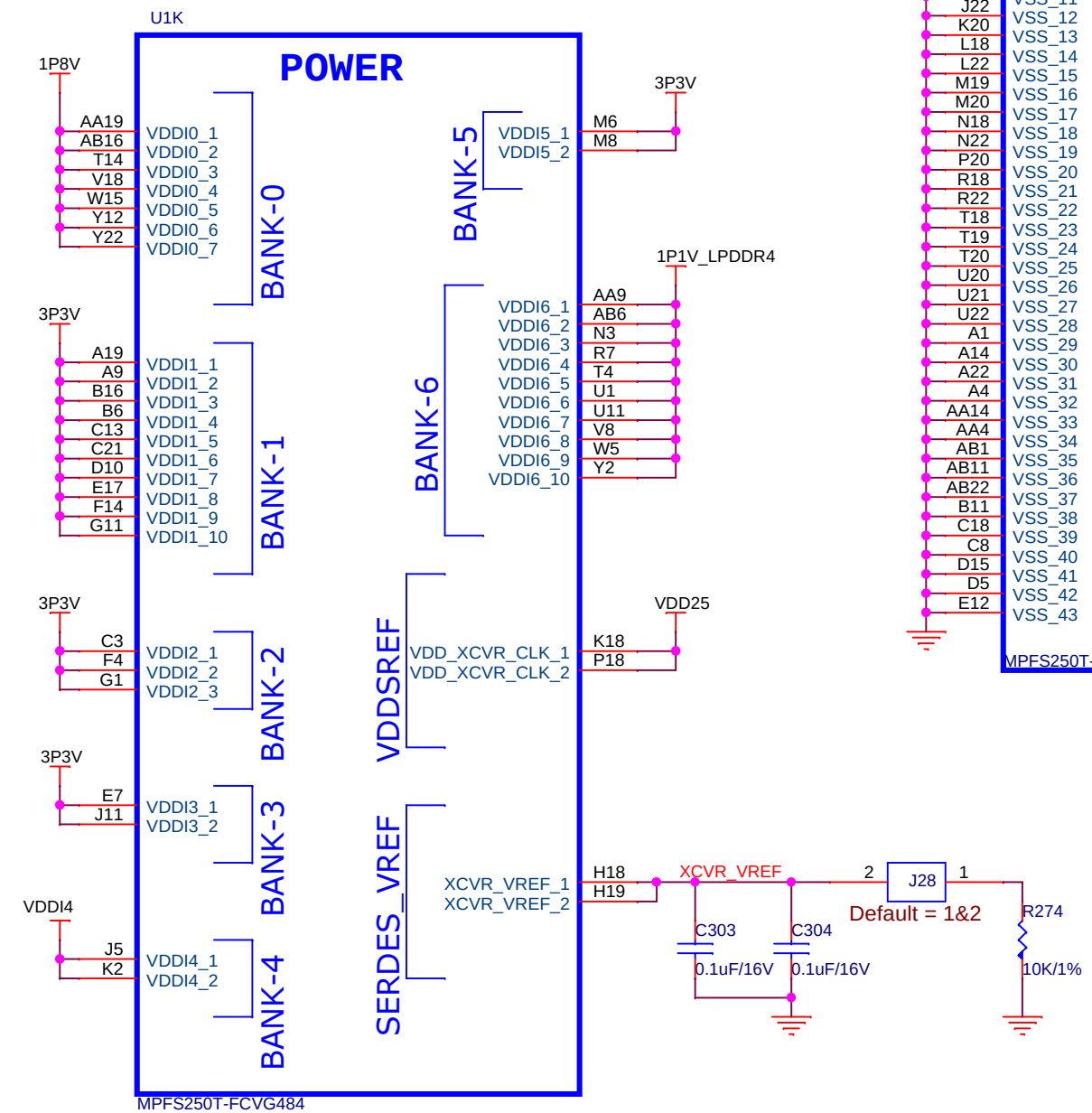
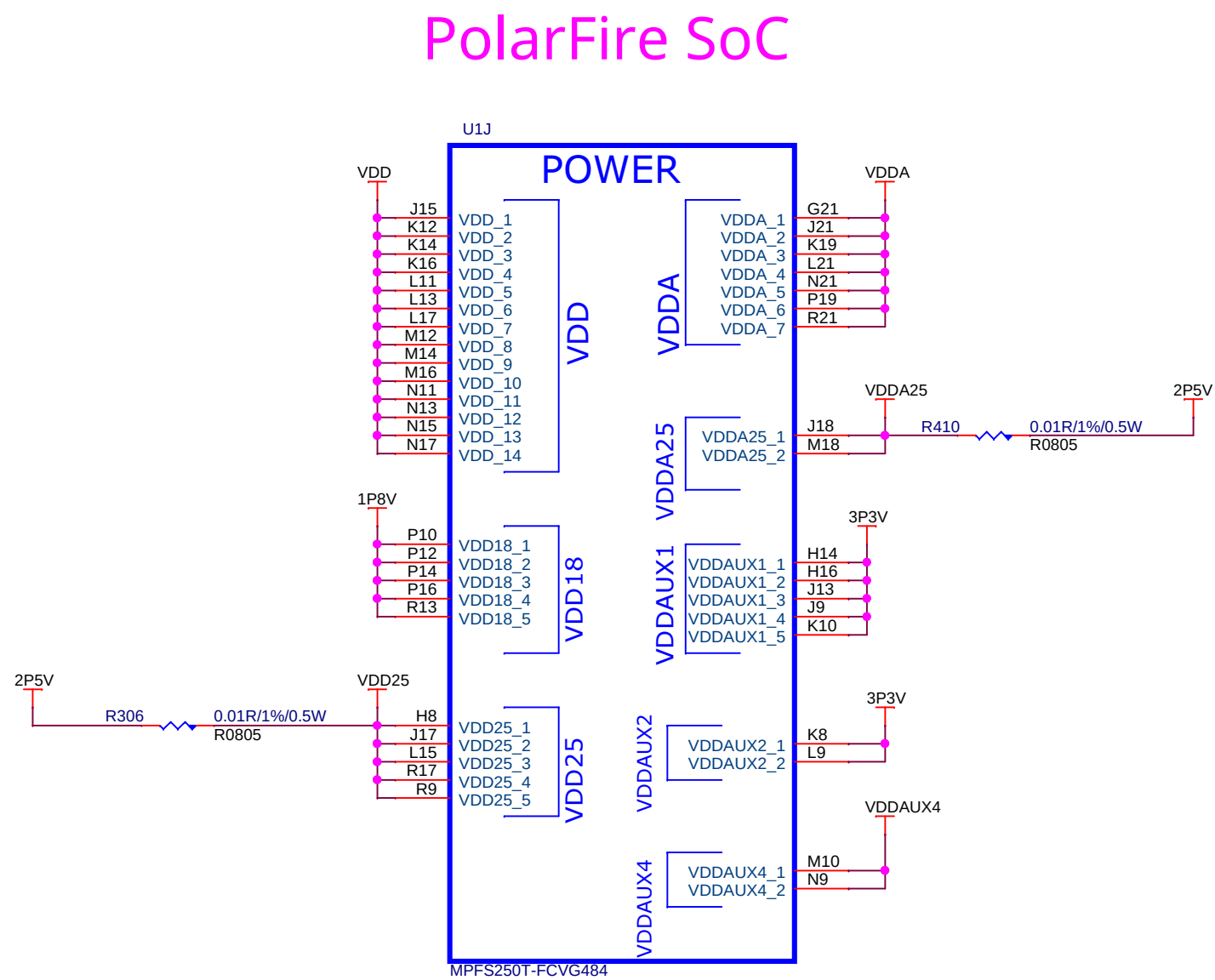


User Defined LEDs



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DATE: Tuesday, June 23, 2020	SH 14 OF 29	

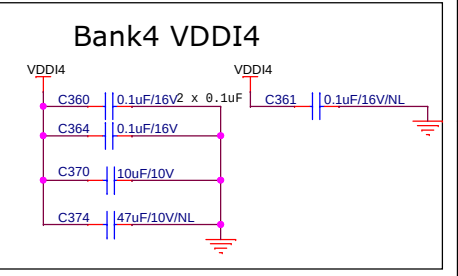
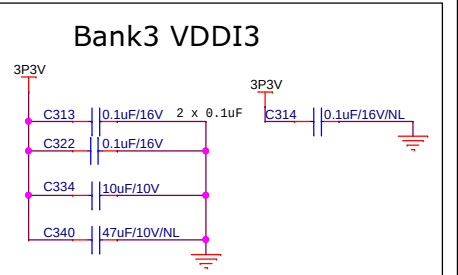
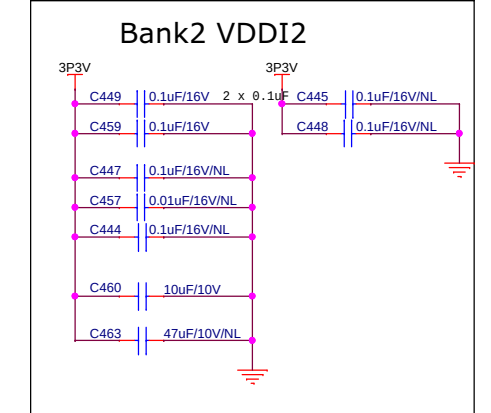
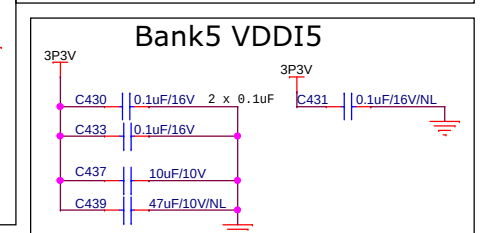
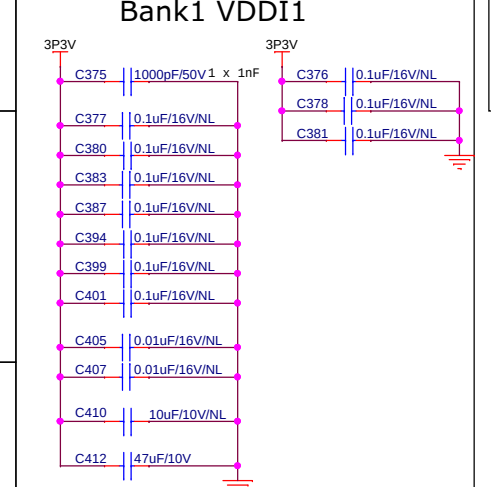
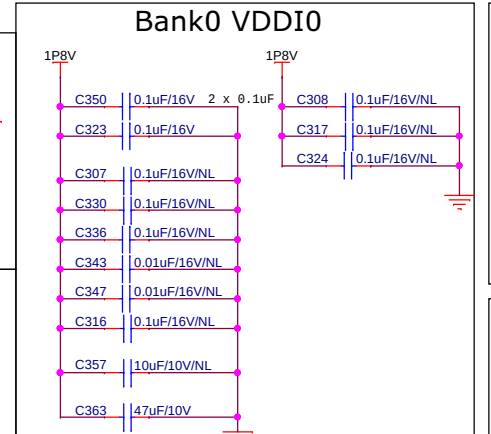
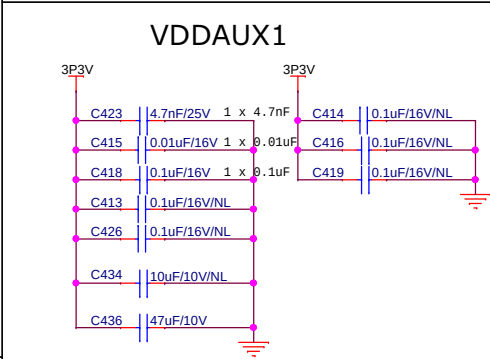
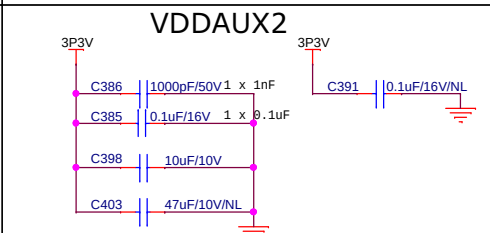
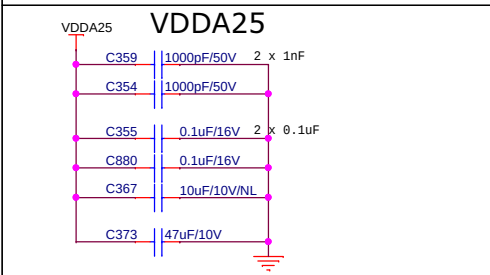
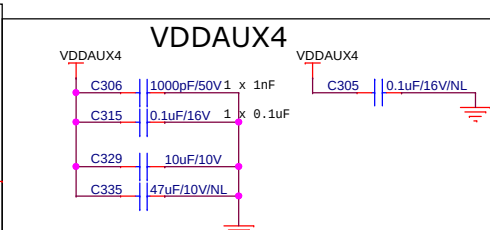
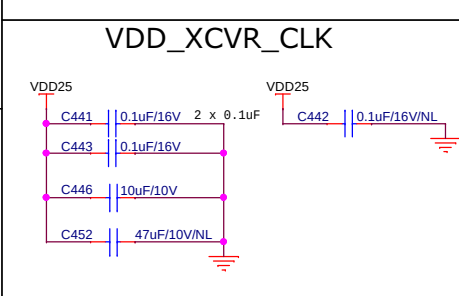
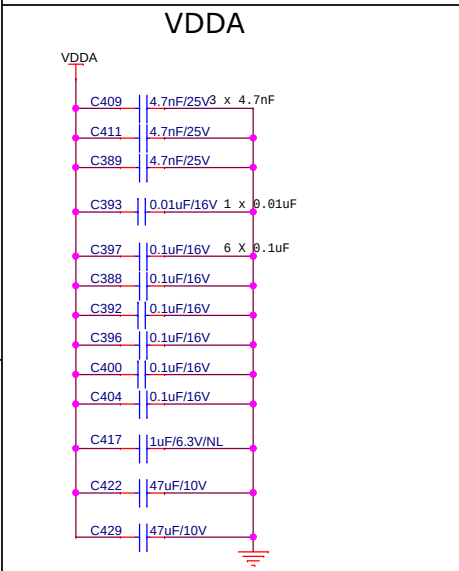
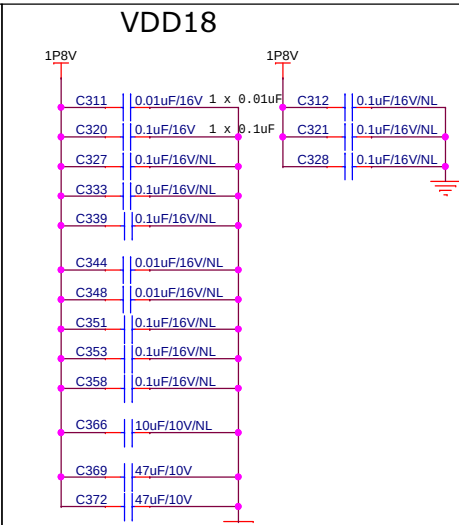
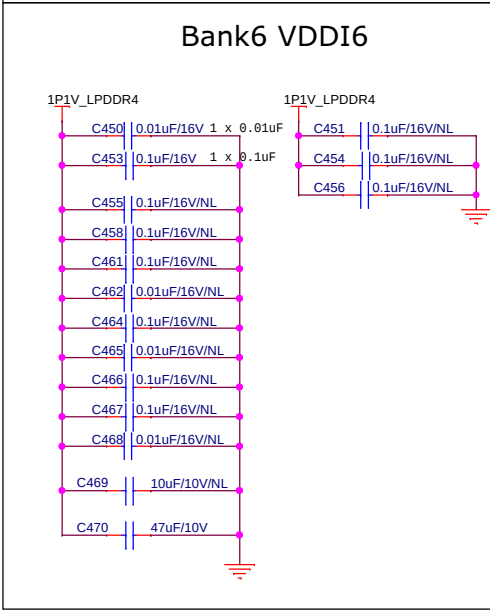
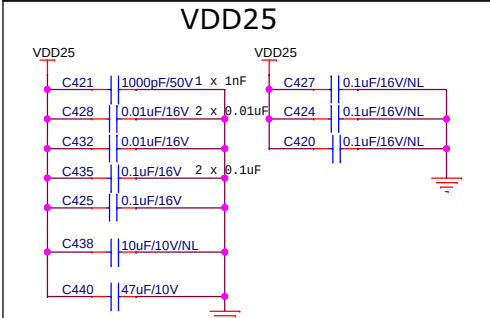
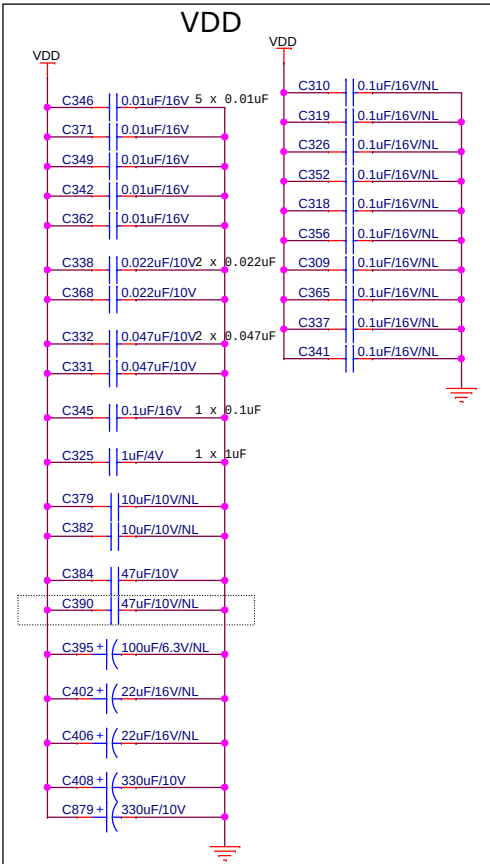
Power Connection



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DATE: Tuesday, June 23, 2020		SH 15 OF 29

DECAPS VALUE TO BE UPDATED

Decoupling Capacitors



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SIZE	DOCUMENT NO.	REV
Custom		1.0
DATE:	Tuesday, June 23, 2020	SH 16 OF 29

BANK0 - DDR3

Smart Fusion 2

U26-1

BANK-0

{18} DDR3_ADDR0
{18} DDR3_ADDR1
{18} DDR3_ADDR2
{18} DDR3_ADDR3
{18} DDR3_ADDR4
{18} DDR3_ADDR5
{18} DDR3_ADDR6
{18} DDR3_ADDR7
{18} DDR3_ADDR8
{18} DDR3_ADDR9
{18} DDR3_ADDR10
{18} DDR3_ADDR11
{18} DDR3_ADDR12
{18} DDR3_ADDR13
{18} DDR3_ADDR14

C14
E11
E12
B16
C16
D16
E16
F15
F14
E15
E14
F13
D14
D13
C15
B15

DDRIO43NB0/MDDR_ADDR0
DDRIO42PB0/MDDR_ADDR1
DDRIO42NB0/MDDR_ADDR2
DDRIO41PB0/MDDR_ADDR3
DDRIO41NB0/MDDR_ADDR4
DDRIO40PB0/MDDR_ADDR5
DDRIO40NB0/MDDR_ADDR6
DDRIO39NB0/MDDR_ADDR7
DDRIO38PB0/MDDR_ADDR8
DDRIO38NB0/MDDR_ADDR9
DDRIO37PB0/MDDR_ADDR10
DDRIO37NB0/MDDR_ADDR11
DDRIO36PB0/MDDR_ADDR12
DDRIO36NB0/MDDR_ADDR13
DDRIO35PB0/MDDR_ADDR14
DDRIO35NB0/MDDR_ADDR15

DDRIO53PB0/MDDR_DQ10/CCC_NE0_CLKI2
DDRIO53NB0/MDDR_DQ11
DDRIO50PB0/MDDR_DQ12/GB12/CCC_NE1_CLKI2
DDRIO50NB0/MDDR_DQ13
DDRIO49PB0/MDDR_DQ14/CCC_NE1_CLKI3
DDRIO49NB0/MDDR_DQ15

DDRIO52PB0/MDDR_DQS1/GB8/CCC_NE0_CLKI3
DDRIO52NB0/MDDR_DQS1_N

DDRIO60PB0/MDDR_DQ0
DDRIO60NB0/MDDR_DQ1
DDRIO59PB0/MDDR_DQ2
DDRIO59NB0/MDDR_DQ3
DDRIO57NB0/MDDR_DQ4
DDRIO56PB0/MDDR_DQ5
DDRIO56NB0/MDDR_DQ6
DDRIO55PB0/MDDR_DQ7
DDRIO54PB0/MDDR_DQ8
DDRIO54NB0/MDDR_DQ9

B2
A2
B6
B5
B3
B7
C6
A7
C8
C9
D7
D8
B11
B10
D9
E8

DDR3_DQ3 {18}
DDR3_DQ1 {18}
DDR3_DQ0 {18}
DDR3_DQ2 {18}
DDR3_DQ5 {18}
DDR3_DQ4 {18}
DDR3_DQ7 {18}
DDR3_DQ6 {18}
DDR3_DQ8 {18}
DDR3_DQ9 {18}
DDR3_DQ10 {18}
DDR3_DQ11 {18}
DDR3_DQ15 {18}
DDR3_DQ13 {18}
DDR3_DQ12 {18}
DDR3_DQ14 {18}

{18} DDR3_DQS0_P
{18} DDR3_DQS0_N

A5
A4

DDRIO58PB0/MDDR_DQS0
DDRIO58NB0/MDDR_DQS0_N

DDRIO52PB0/MDDR_DQS1/GB8/CCC_NE0_CLKI3
DDRIO52NB0/MDDR_DQS1_N

A9
A10

DDR3_DQS1_P {18}
DDR3_DQS1_N {18}

{18} DDR3_WE#
{18} DDR3_RAS#
{18} DDR3_CAS#
{18} DDR3_CKE0
{18} DDR3_CS0

C11
B12
A12
E10
D11
A13
F16

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DDRIO48PB0/MDDR_RAS_N
DDRIO46NB0/MDDR_CAS_N
DDRIO47PB0/MDDR_CKE
DDRIO47NB0/MDDR_CS_N
DDRIO46PB0/MDDR_RESET_N
DDRIO39PB0/MDDR_ODT

DDRIO44PB0/MDDR_BA0
DDRIO44NB0/MDDR_BA1
DDRIO43PB0/MDDR_BA2

C12
D12
B13

DDR3_BA0 {18}
DDR3_BA1 {18}
DDR3_BA2 {18}

DDRIO45PB0/MDDR_CLK
DDRIO45NB0/MDDR_CLK_N

A14
A15

DDR3_CLK0_P {18}
DDR3_CLK0_N {18}

{18} DDR3_ODT0

B8
A8

DDRIO51PB0/MDDR_TMATCH_0_IN
DDRIO55NB0/MDDR_TMATCH_0_OUT

DDRIO63PB0/MDDR_TMATCH_ECC_IN
DDRIO66PB0/MDDR_TMATCH_ECC_OUT

C4
D3

DDR3_TMATCH_ECC_IO

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{18} DDR3_DQM1

A3
C10

DDRIO57PB0/MDDR_DM_RDQS0
DDRIO51NB0/MDDR_DM_RDQS1

DDRIO61NB0
DDRIO65PB0/GB0/CCC_NW0_CLKI3
DDRIO65NB0/GB4/CCC_NW1_CLKI2
DDRIO63NB0/MDDR_DM_RDQS_ECC

D2
D4
E4
C3

D6
C5

DDRIO62NB0/MDDR_DQ_ECC2
DDRIO62PB0/MDDR_DQ_ECC3

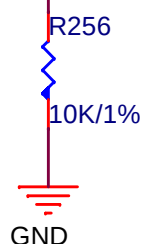
C1
B1

DDRIO64PB0/MDDR_DQS_ECC
DDRIO64NB0/MDDR_DQS_ECC_N

D1
E2

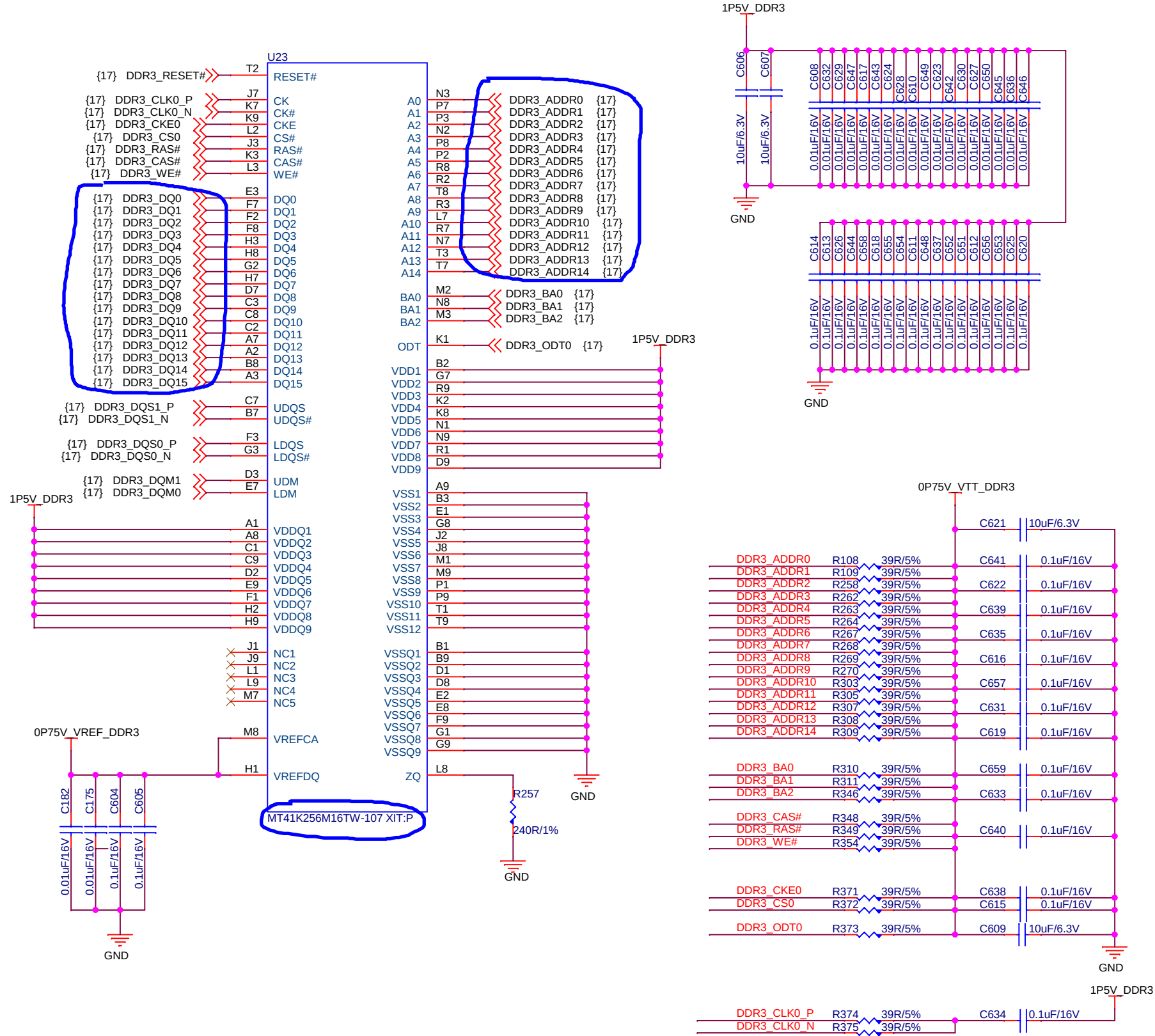
DDRIO61PB0/CCC_NW1_CLKI3
DDRIO66NB0/CCC_NW0_CLKI2

M2S025-VFG256



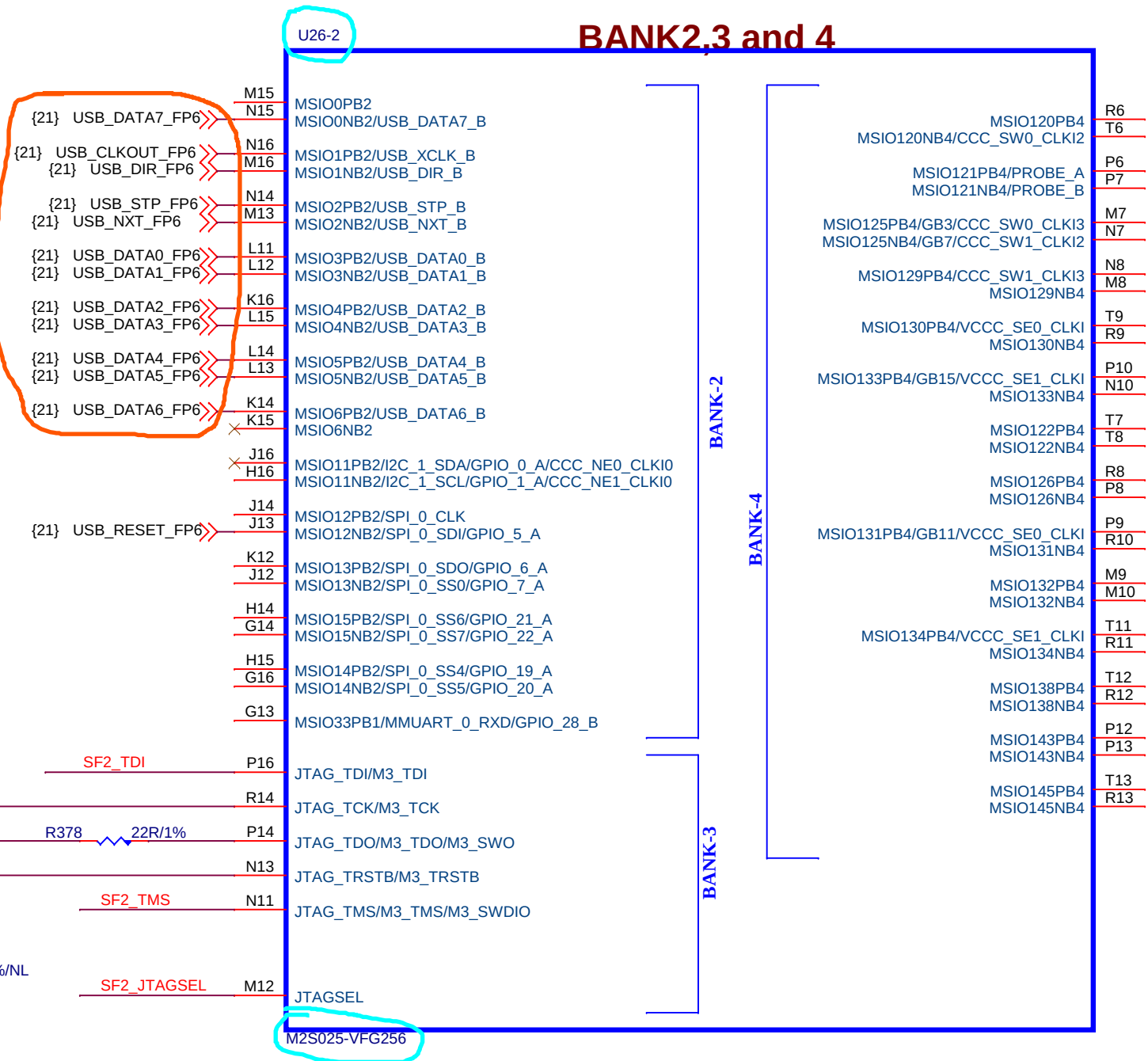
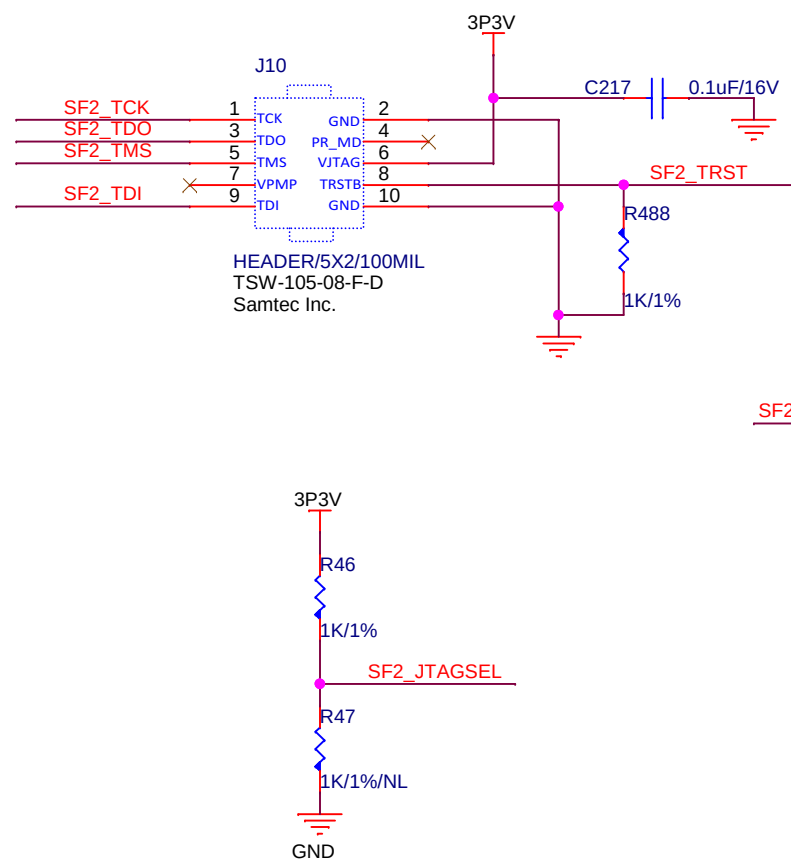
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DATE:	Tuesday, June 23, 2020	SH 17 OF 29

DDR3 MEMORY -BANK0



SmartFusion2

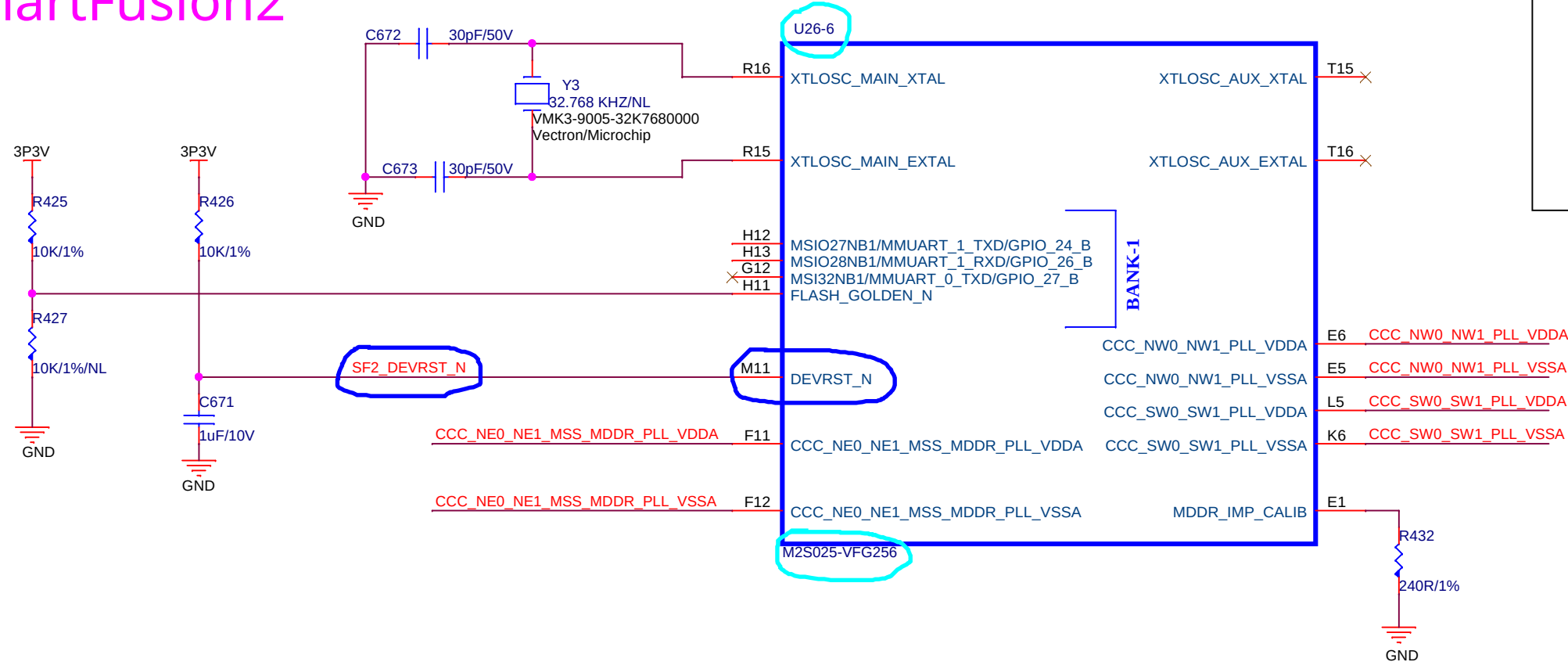
Programming connector for SF2



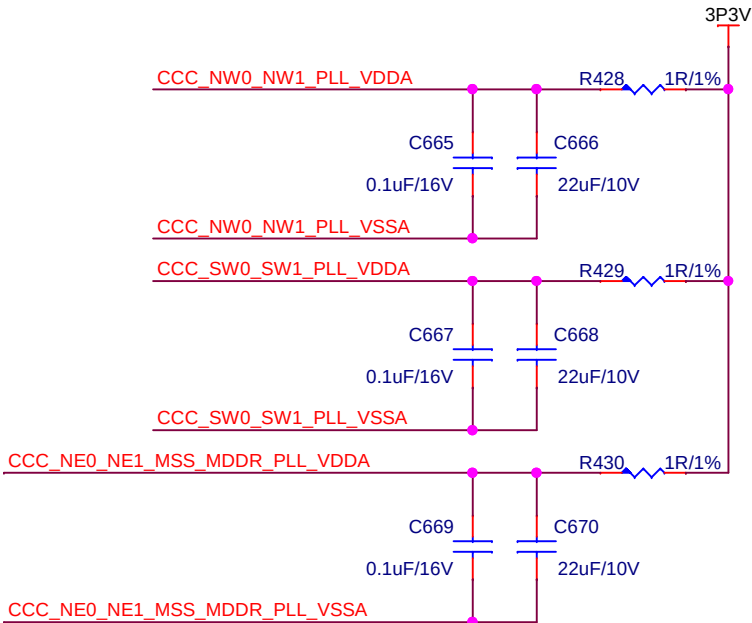
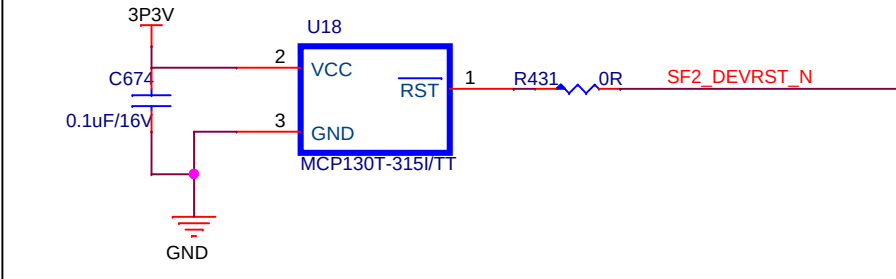
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SIZE	DOCUMENT NO.	REV
Custom		1.0
DATE:	Tuesday, June 30, 2020	SH 19 OF 29

SmartFusion2

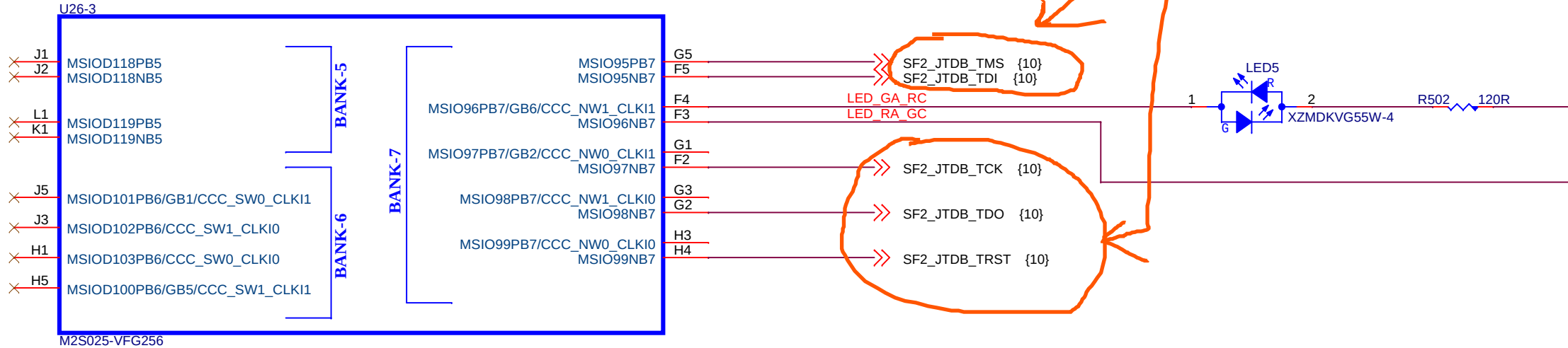
BANK-1,5,6,7 and MISC



DEVICE_RESET



Use SmartFusion2 to program PolarFire-SoC

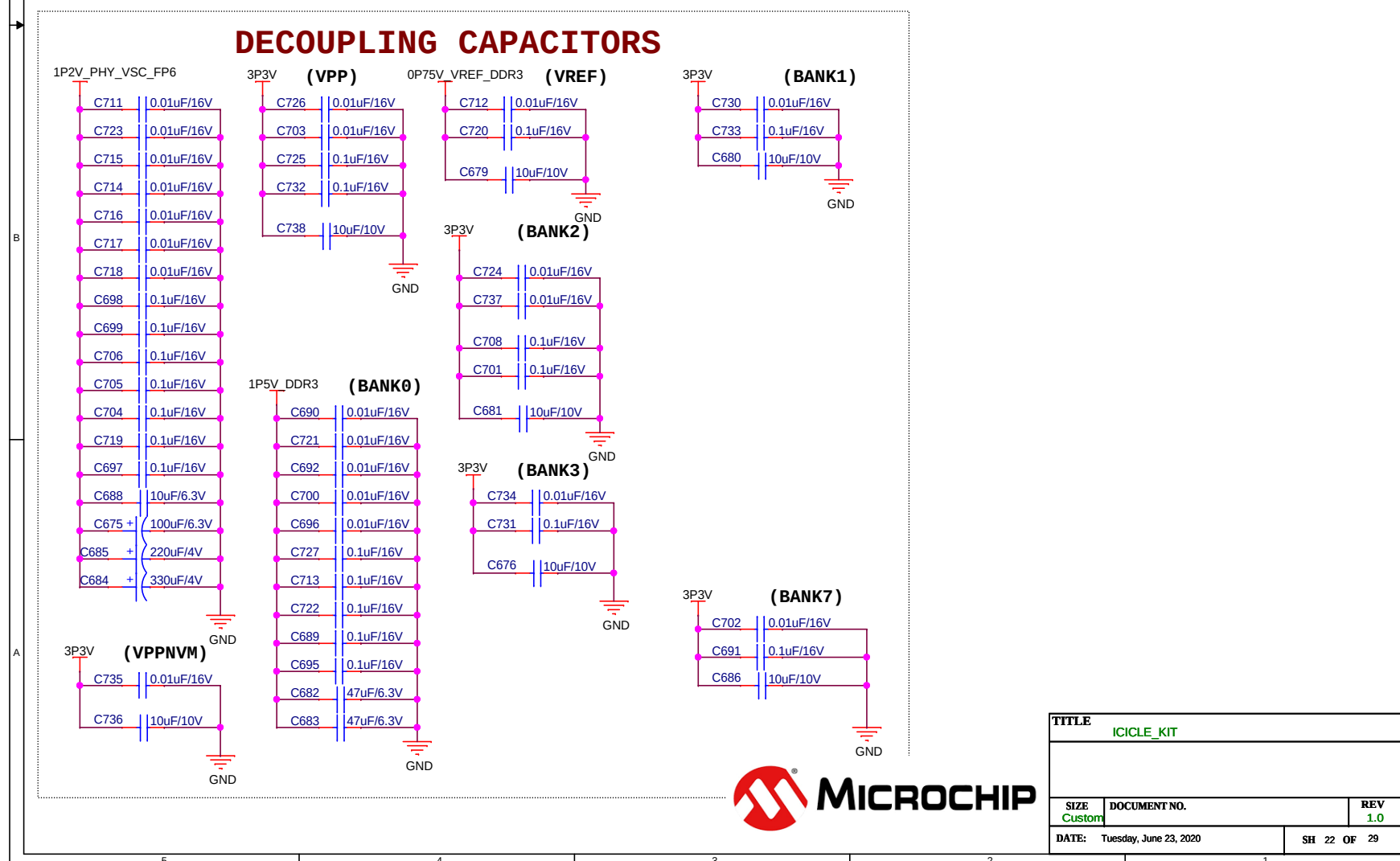
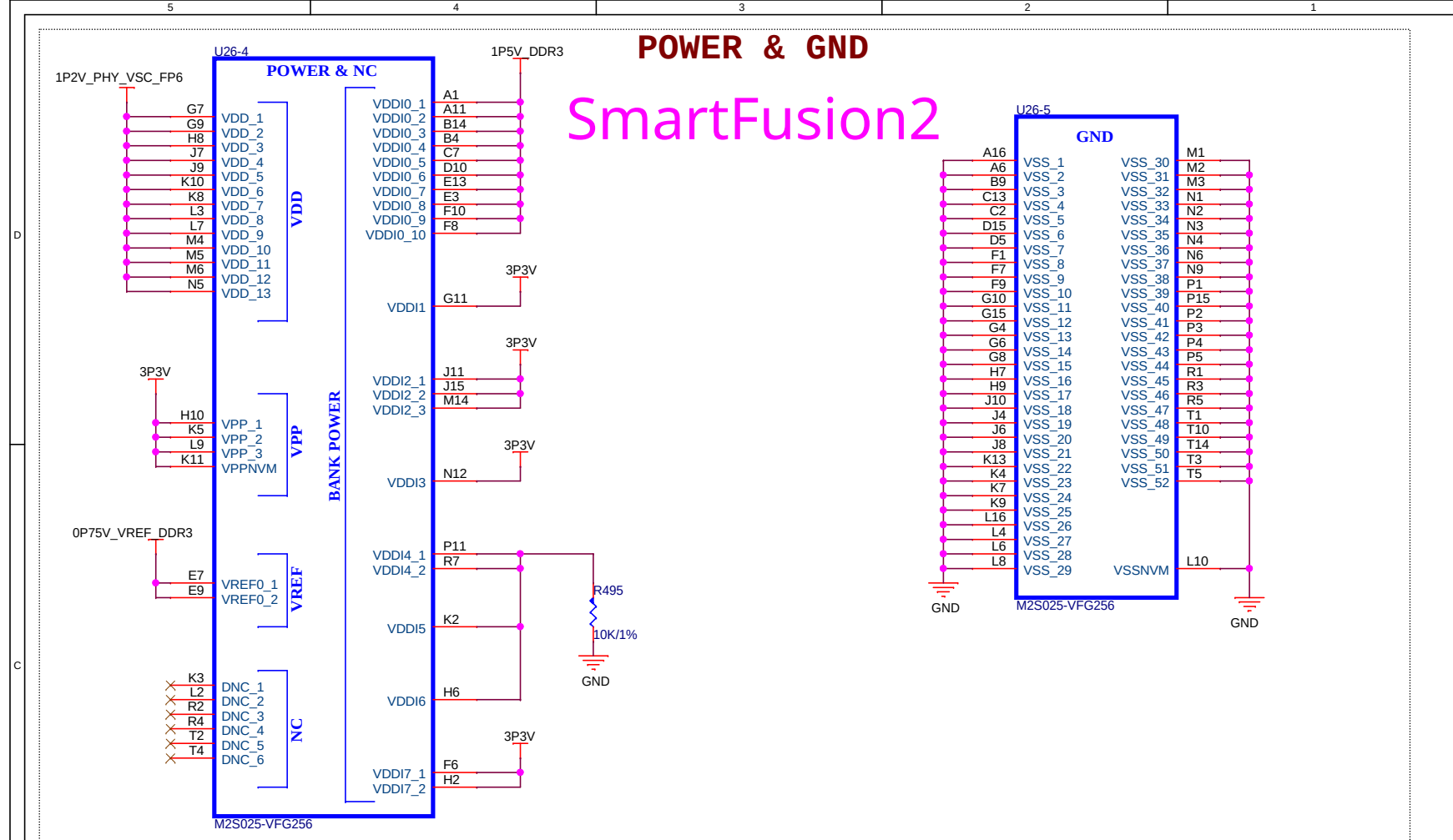


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DATE:	Tuesday, June 23, 2020	SH 20 OF 29

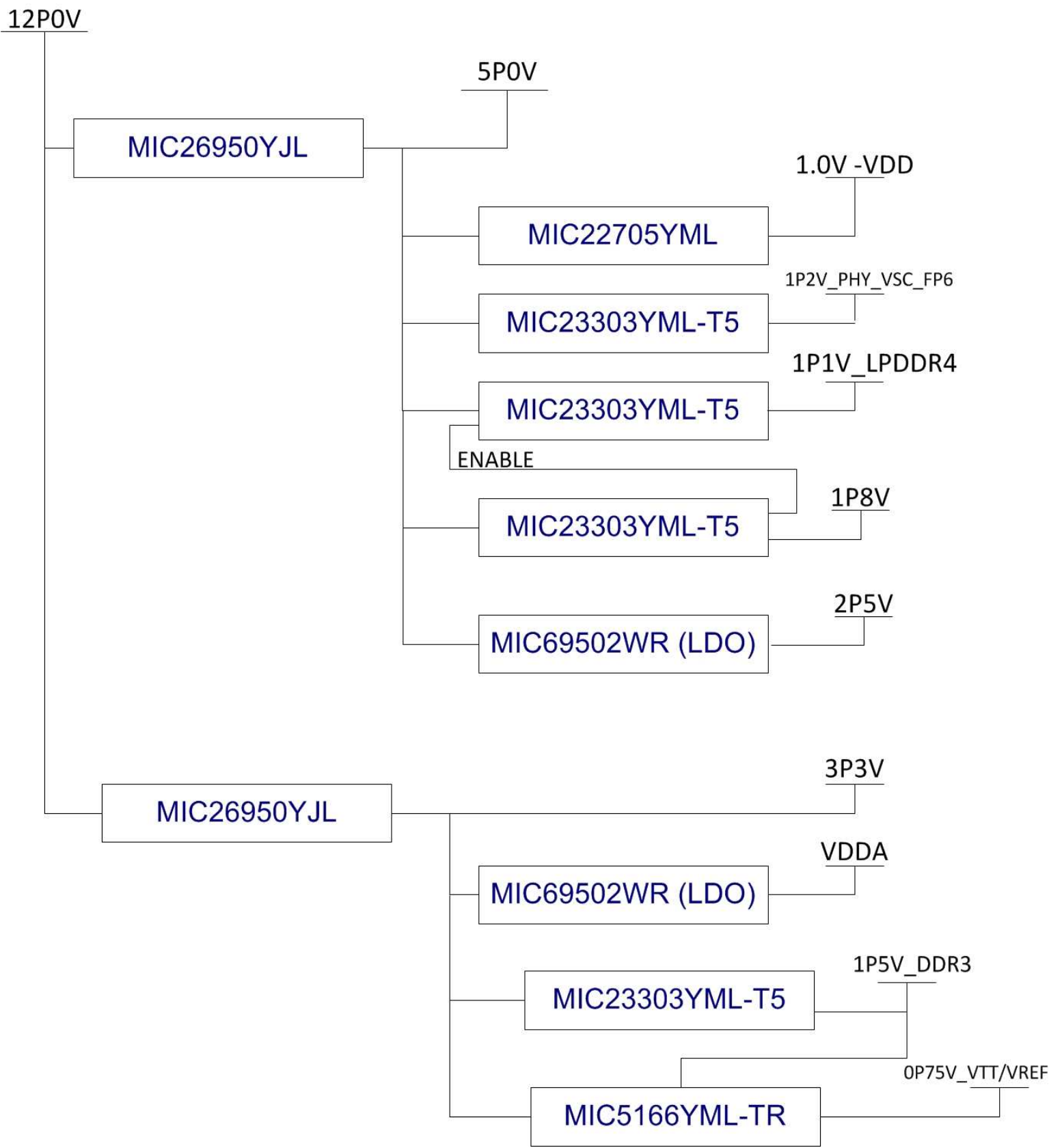
Schematic shows 26.0 MHz Configuration.
REFCLK[2:0] = 110



TITLE			
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SIZE	DOCUMENT NO.		REV
Custom			1.0
DATE: Tuesday, June 23, 2020			SH 21 OF 29

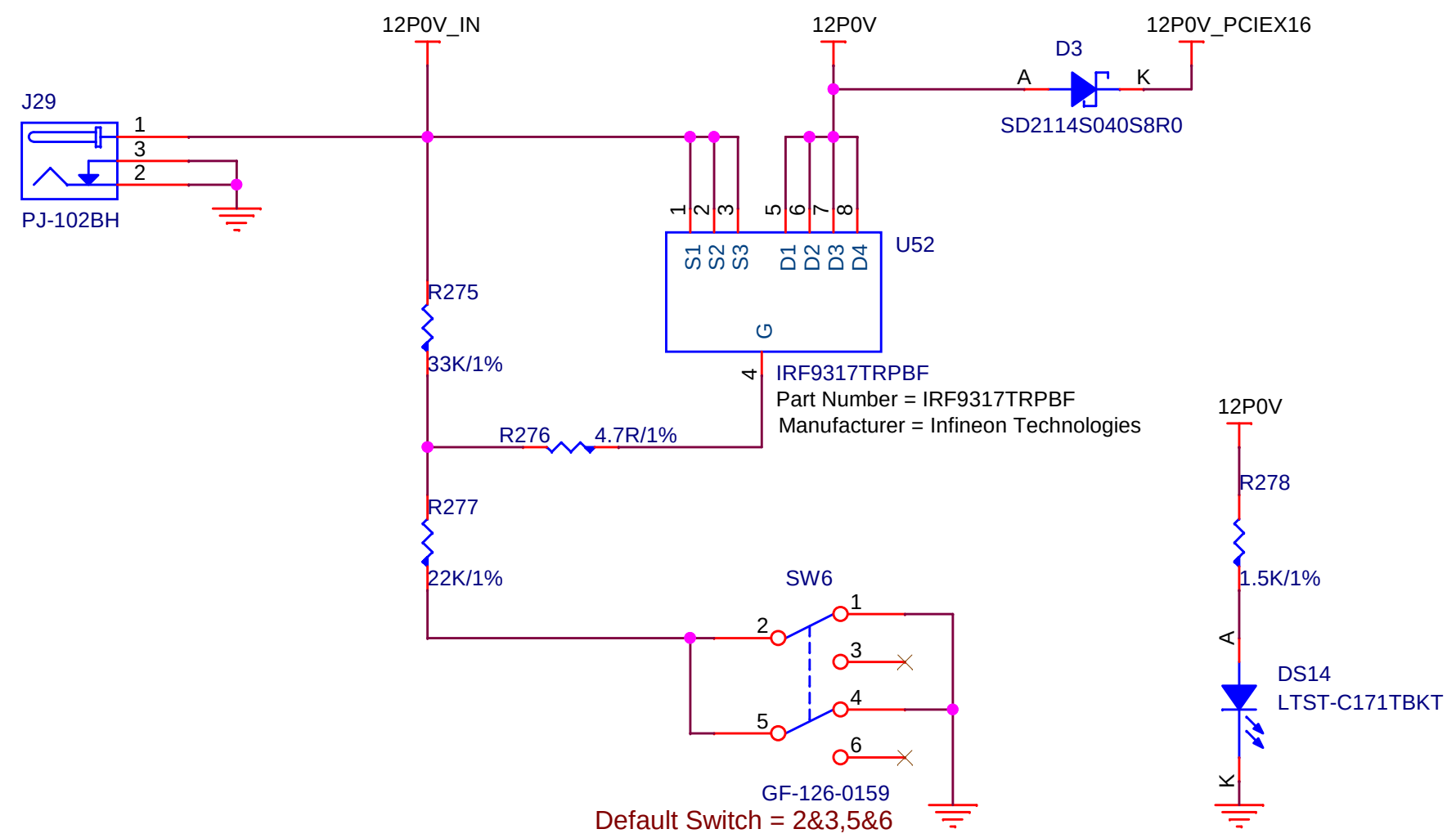


POWER BLOCK DIAGRAM



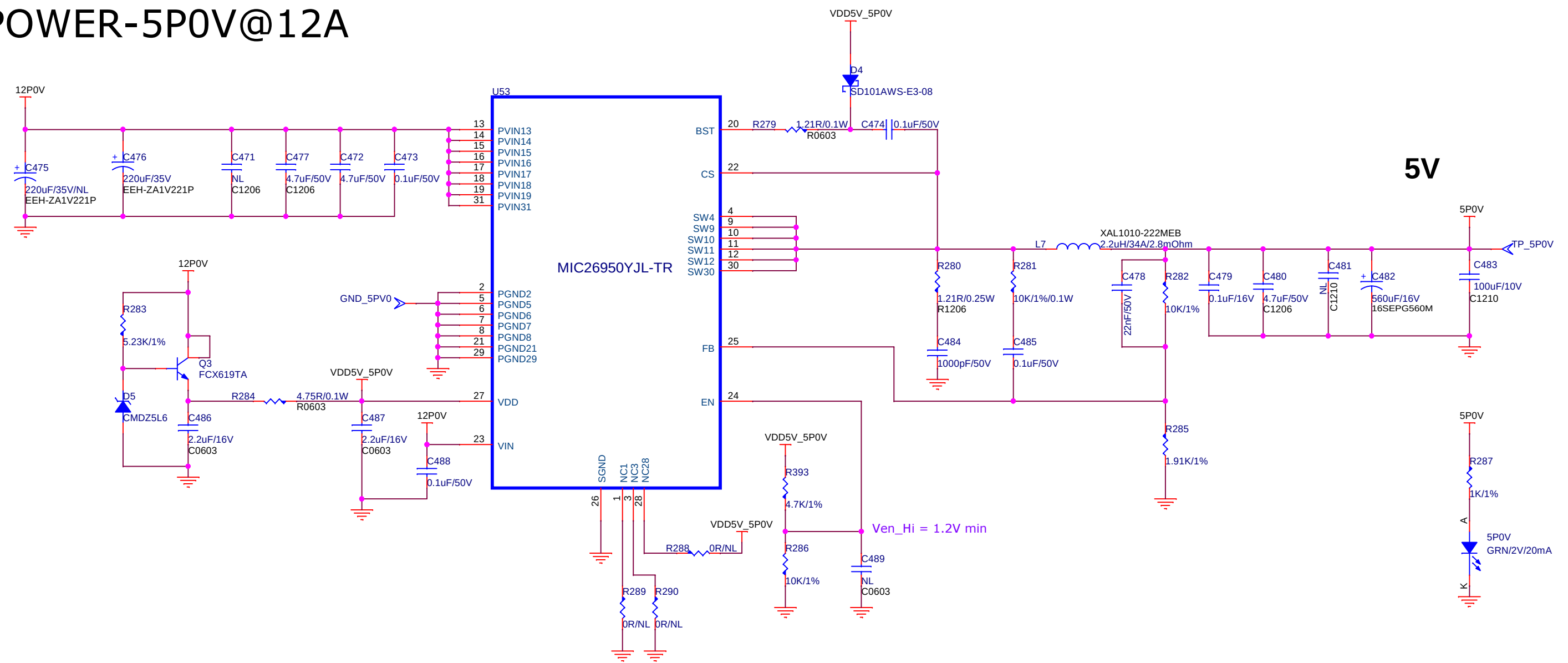
TITLE		
ICICLE_KIT		
SIZE B	DOCUMENT NO.	REV 1.0
DATE: Tuesday, June 23, 2020	SH 23 OF 29	

12V External Supply @ 5A

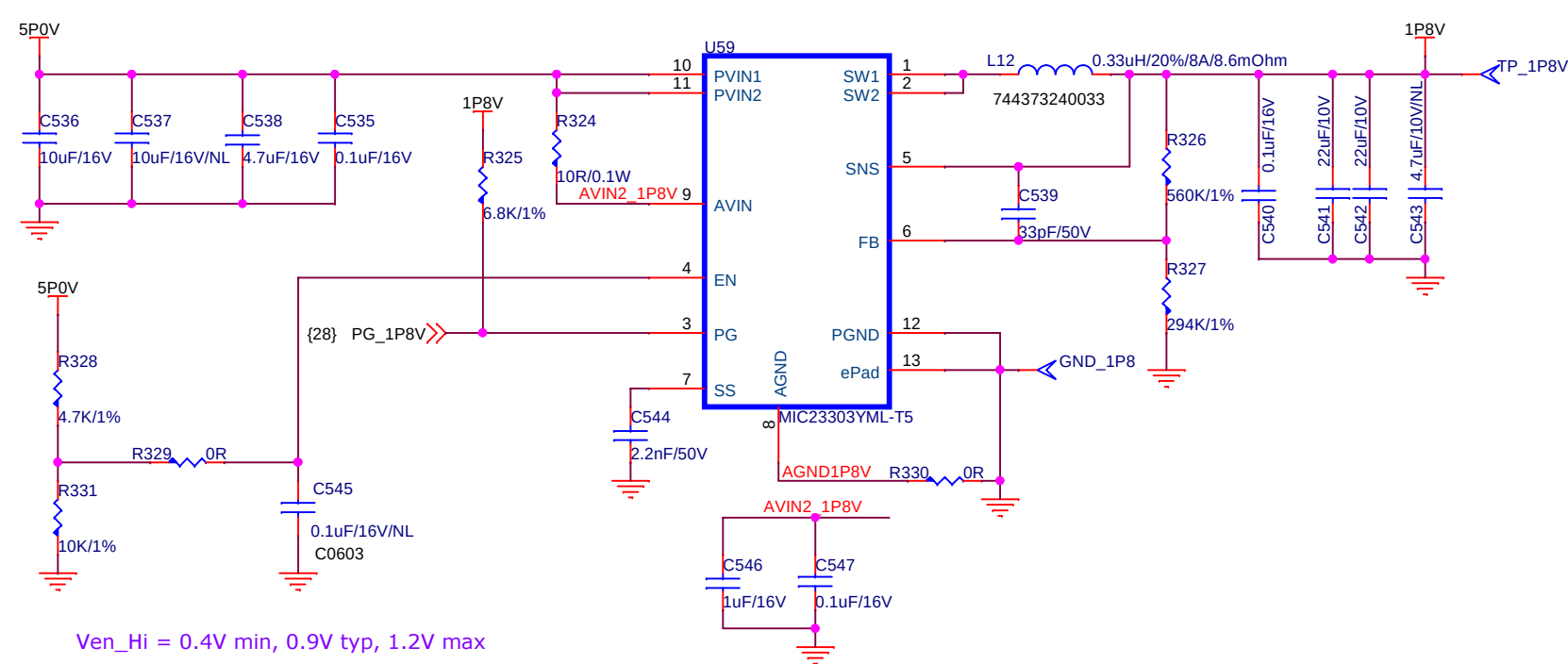


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DATE:	Tuesday, June 23, 2020	SH 24 OF 29

POWER-5P0V@12A

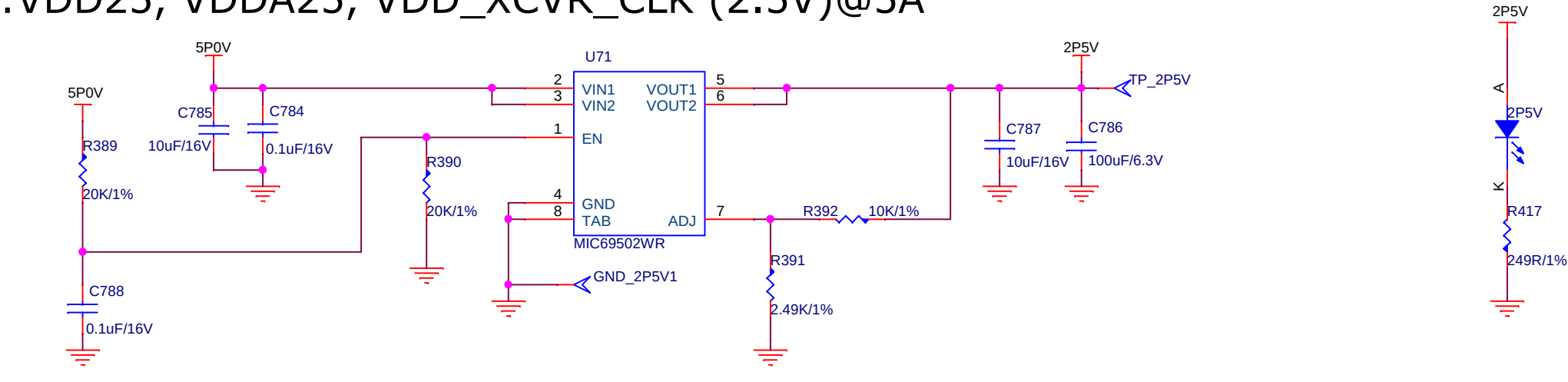


POWER:1P8V(1.8V)@3A

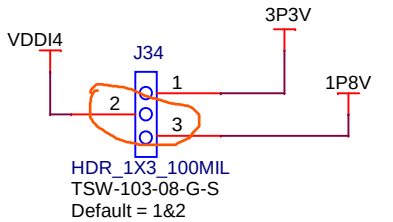


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ICICLE_KIT		
SIZE	DOCUMENT NO.	REV
Custom		1.0
DATE:	Tuesday, June 23, 2020	SH 25 OF 29

POWER:VDD25, VDDA25, VDD_XCVR_CLK (2.5V)@5A

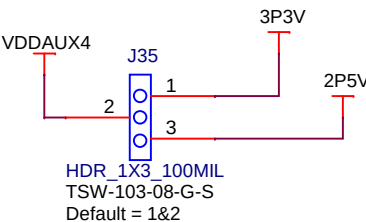


VDDI4



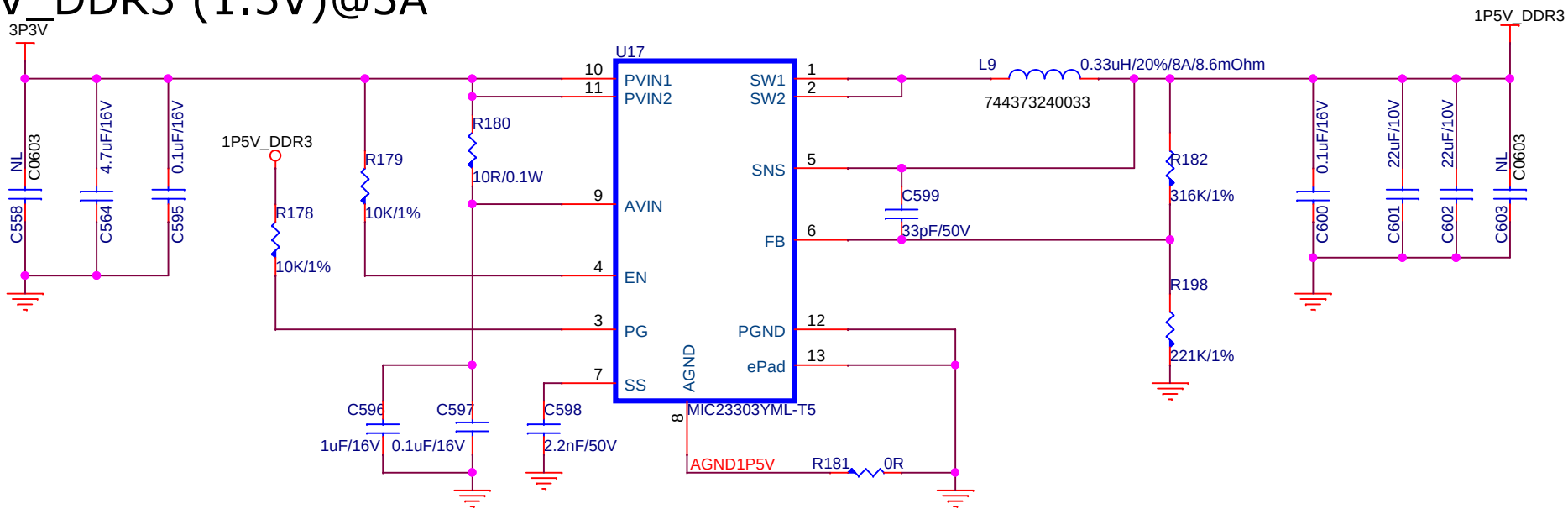
6A Capable header

VDDAUX4

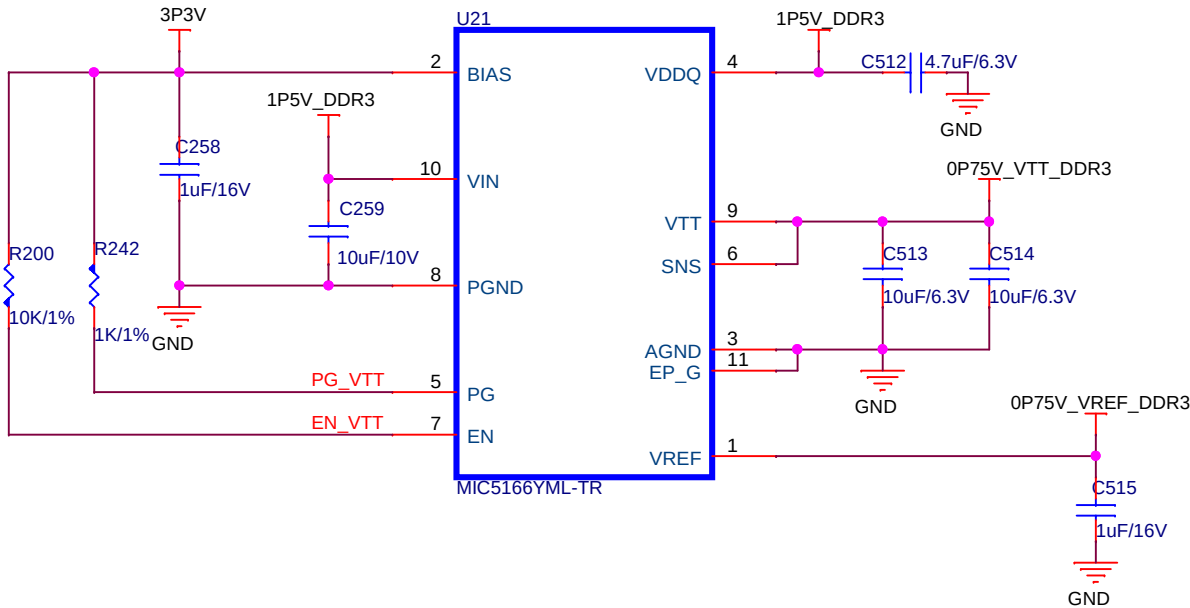


6A Capable header

POWER:1P5V_DDR3 (1.5V)@3A

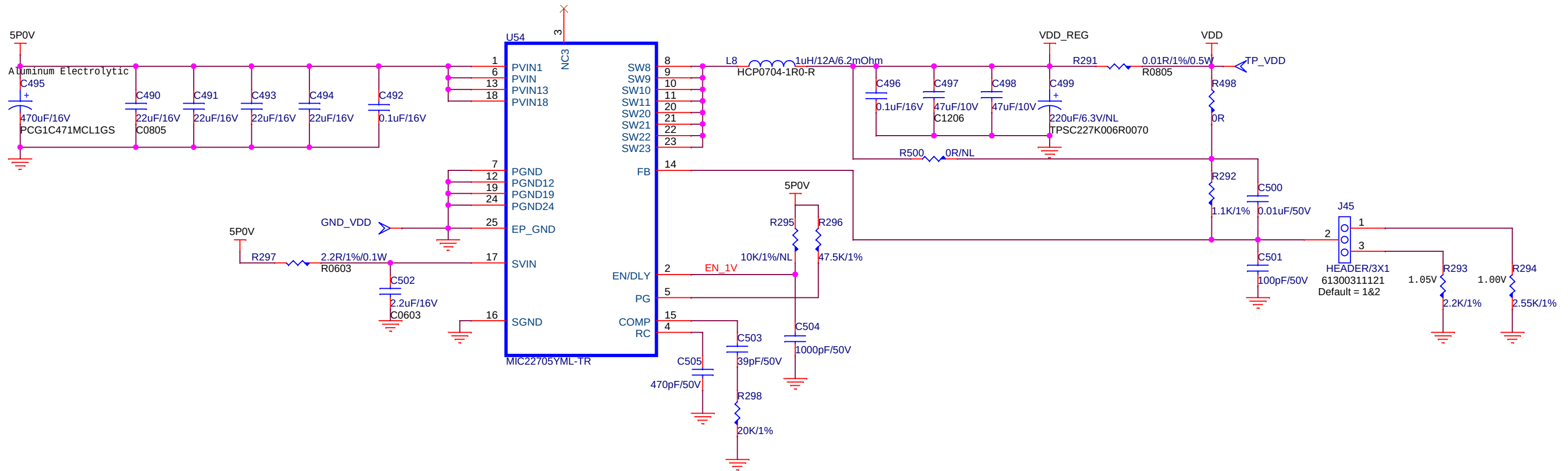


POWER:DDR3 VTT & VREF

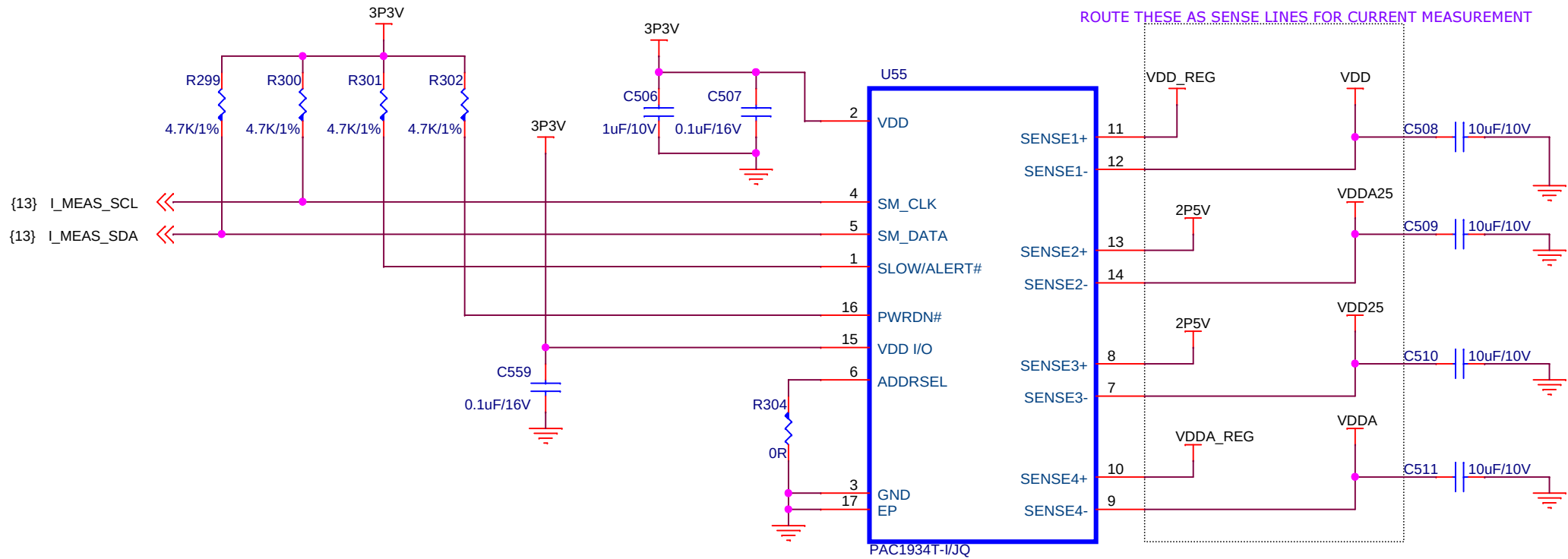


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ICICLE_KIT		
SIZE	DOCUMENT NO.	REV
B		1.0
DATE:	Tuesday, June 23, 2020	SH 26 OF 29

POWER:VDD(1.0/1.05V)@7A



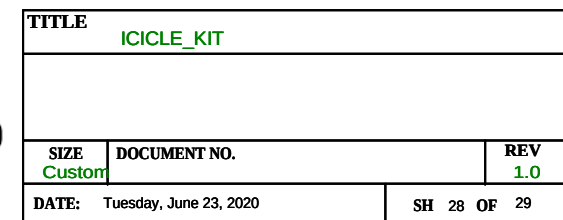
Current Measurement 1



I2C Address - 0010_000(R/W)



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SIZE B	DOCUMENT NO.	REV 1.0
DATE: Tuesday, June 23, 2020		SH 27 OF 29

[illegible]

POWER: 3P3V (3.3V)@12A

The schematic shows a DC-DC converter circuit using the MIC26950YJL-TR. The input is 12V, and the output is 3.3V. The circuit includes a MOSFET (Q4), an inductor (L11), and various capacitors and resistors for filtering and compensation. The output is 3.3V at 12A. A note specifies $V_{en_Hi} = 1.2V \text{ min}$.

The schematic diagram illustrates the MIC69502WR voltage converter circuit. The input is a 3P3V supply connected to the VIN1 pin (pin 2) of the MIC69502WR (U61). A 4.7K/1% resistor (R338) and a 0.33uF/10V capacitor (C557) are connected in parallel between the input and ground. The EN pin (pin 1) is connected to the input supply through a 10uF/10V capacitor (C553) and a 0.1uF/16V capacitor (C554) in parallel. The GND pin (pin 4) is connected to ground through a 4.7K/1% resistor (R339). The VOUT1 pin (pin 5) is connected to the VDDA_REG supply. The VOUT2 pin (pin 6) is connected to the VDDA supply. The ADJ pin (pin 7) is connected to ground. The VDDA supply is regulated by a 0.01R/1%/0.5W resistor (R0805) and a 10uF/16V capacitor (C555) in parallel. The VDDA supply is also connected to the VDDA pin of the MIC69502WR (U61) through a 100uF/6.3V capacitor (C556). The VDDA supply is connected to the VDDA pin of the MIC69502WR (U61) through a 10K/1% resistor (R340) and a 9.09K/1% resistor (R341) in series. The VDDA supply is also connected to the VDDA pin of the MIC69502WR (U61) through a 0R resistor (R499).

